

Data Analytics Assignment Report



Submitted

by

NAME	PRN	ROLL NO
SOURAV SUMAN	2314110523	83

Submitted To:

Prof. SWAPNALI MAHADIK

Colab Notebook:

ASSIGNMENT1:-

<https://colab.research.google.com/drive/157O99ZYtR4uFgRoAdOcbkEZgiRpMkBth?usp=sharing>

ASSIGNMENT2:-https://colab.research.google.com/drive/1szbUIrya42yP-jTuLPcJLhm_jzIP7Vp4?usp=sharing

ASSIGNMENT3:-

<https://colab.research.google.com/drive/1AIDmsXjo35dPCS3AuQZA397vHeVQeJdg?usp=sharing>

ASSIGNMENT4:-

<https://colab.research.google.com/drive/1i9ZhVnz6NUgthtDbTporzmEYuPfgrzWq?usp=sharing>

ASSIGNMENT5:-

https://colab.research.google.com/drive/1Qvluui_o9K_WHOYnZGZ51hNH0sIdF_8?usp=sharing

ASSIGNMENT6:-[https://colab.research.google.com/drive/1Pe2ItQaJ93cPwk377q4m3Uk-](https://colab.research.google.com/drive/1Pe2ItQaJ93cPwk377q4m3Uk-5k3JfwLD?usp=sharing)

[5k3JfwLD?usp=sharing](https://colab.research.google.com/drive/1Pe2ItQaJ93cPwk377q4m3Uk-5k3JfwLD?usp=sharing)

Github Link:-https://github.com/LEVELING2108/DATA_ANALYTICS.git

Department of Electronics & Communication Engineering
Bharati Vidyapeeth (Deemed to be University)
College of Engineering,
Pune – 411043

Academic Year: 2025–26

Data Analytics Assignment

Question 1-Solution

Understanding Data & Attribute Classification

Dataset: insurance.csv

Records: 1,338 (1,337 after cleaning)

Attributes: age, sex, bmi, children, smoker, region, charges

Google Colab -[LINK](#)

Part A – Attribute Type Identification

#	Attribute	Type	Justification
1	age	Discrete	Integer count of years; recorded as whole numbers only (18–64)
2	sex	Nominal	Two categories (male/female) with no inherent ordering.
3	bmi	Continuous	Real-valued measurement on a continuous scale (e.g., 27.9, 33.77, 22.705)
4	children	Discrete	Countable integer values representing number of children (0, 1, 2, ..., 5).
5	smoker	Nominal	Binary categories (yes/no) with no ordinal relationship.
6	region	Nominal	Four geographic labels (northeast, northwest, southeast, southwest) with no natural order.
7	charges	Continuous	Monetary amount that can take any real value within a range (e.g., 1121.87 – 63770.43).

#Summary: 2 Nominal · 2 Discrete · 2 Continuous · 0 Ordinal

Part B – Data Extraction & Cleaning

Task 1 — Load Dataset, Display First 10 Rows & Info:-

Code:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
# Load dataset from Google Drive url =
"https://drive.google.com/uc?export=download&id=1oyN6CXzbJq42dL5Jqkn1cP83Hu93CD6q"
df = pd.read_csv(url)
# Display first 10 rows
print(df.head(10))
# Display dataset info
df.info()

```

Output — First 10 Rows:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.9240
1	18	male	33.770	1	no	southeast	1725.5523
2	28	male	33.000	3	no	southeast	4449.4620
3	33	male	22.705	0	no	northwest	21984.4706
4	32	male	28.880	0	no	northwest	3866.8552
5	31	female	25.740	0	no	southeast	3756.6216
6	46	female	33.440	1	no	southeast	8240.5896
7	37	female	27.740	3	no	northwest	7281.5056
8	37	male	29.830	2	no	northeast	6406.4107
9	60	female	25.840	0	no	northwest	28923.1369

Output — Dataset Info:

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column    Non-Null Count  Dtype
---  -
0  age        1338 non-null   int64

```

```
1 sex          1338 non-null  object
2 bmi          1338 non-null  float64
3 children     1338 non-null  int64
4 smoker       1338 non-null  object
5 region       1338 non-null  object
6 charges      1338 non-null  float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

Task 2 — Missing Values & Duplicate Records:-

Code:--

```
# Check missing values
print(df.isnull().sum())
print(f"\nTotal missing values: {df.isnull().sum().sum()}")

# Check duplicates
duplicates = df.duplicated().sum()
print(f"Duplicate rows: {duplicates}")

# Handle duplicates
if duplicates > 0:
    df = df.drop_duplicates()
    print(f"Removed duplicates. New shape: {df.shape}")
```

Output:-

```
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```

```
Total missing values: 0
Duplicate rows: 1
Removed duplicates. New shape: (1337, 7)
```

Justification:

One duplicate row was found and removed using `df.drop_duplicates()`. Removing duplicates is standard practice because duplicate records artificially inflate sample size and can bias statistical analysis and model training. No missing values were found, so no imputation was needed.

Task 3 — Data Inconsistencies:-

Code:

```
print(f'Negative charges: {(df['charges'] < 0).sum()}')  
print(f'Unrealistic BMI (<10 or >60): {((df['bmi'] < 10) | (df['bmi'] > 60)).sum()}')  
print(f'Negative age: {(df['age'] < 0).sum()}')  
print(f'Negative children: {(df['children'] < 0).sum()}')  
  
display(df.describe())
```

Output:

Negative charges: 0
Unrealistic BMI (<10 or >60): 0
Negative age: 0
Negative children: 0

Descriptive Statistics:

Statistic	age	bmi	children	charges
count	1337.00	1337.00	1337.00	1337.00
mean	39.21	30.66	1.09	13270.42
std	14.05	6.10	1.21	12110.01
min	18.00	15.96	0.00	1121.87
25%	27.00	26.30	0.00	4740.29
50%	39.00	30.40	1.00	9382.03
75%	51.00	34.69	2.00	16639.91
max	64.00	53.13	5.00	63770.43

Conclusion:

No obvious data inconsistencies were detected. All values fall within realistic ranges:

- Age: 18–64 (valid adult range)
- BMI: 15.96–53.13 (within plausible human range)
- Children: 0–5 (reasonable count)
- Charges: all positive (no negative monetary values)

Task 4 — Categorical Encoding:-

Code:

```
# Smoker: binary encoding (yes → 1, no → 0)
df['smoker_encoded'] = df['smoker'].map({'yes': 1, 'no': 0})

# Region: one-hot encoding using get_dummies
df_encoded = pd.get_dummies(df, columns=['region'], prefix='region', dtype=int)

print(df[['smoker', 'smoker_encoded']].head(10))
display(df_encoded[['region_northeast', 'region_northwest', 'region_southeast', 'region_southwest']])
.head(10))
```

Output — Smoker Encoding:

	smoker	Smoker_encoded
0	yes	1
1	no	0
2	no	0
3	no	0
4	no	0
5	no	0

6	no	0
7	no	0
8	no	0
9	no	0

Output — Region One-Hot Encoding (first 10 rows):-

	Region_northeast	Region_northwest	Region_southeast	Region_southwest
0	0	0	0	1
1	0	0	1	0
2	0	0	1	0
3	0	1	0	0
4	0	1	0	0
5	0	0	1	0
6	0	0	1	0
7	0	1	0	0
8	1	0	0	0
9	0	1	0	0

Explanation of Transformations:--

Smoker (Binary Encoding): The smoker column has exactly two categories (yes/no). Mapping yes → 1 and no → 0 creates a single binary numeric column. This is appropriate for binary variables because it preserves all information in minimal space and is directly usable in correlation analysis and regression models.

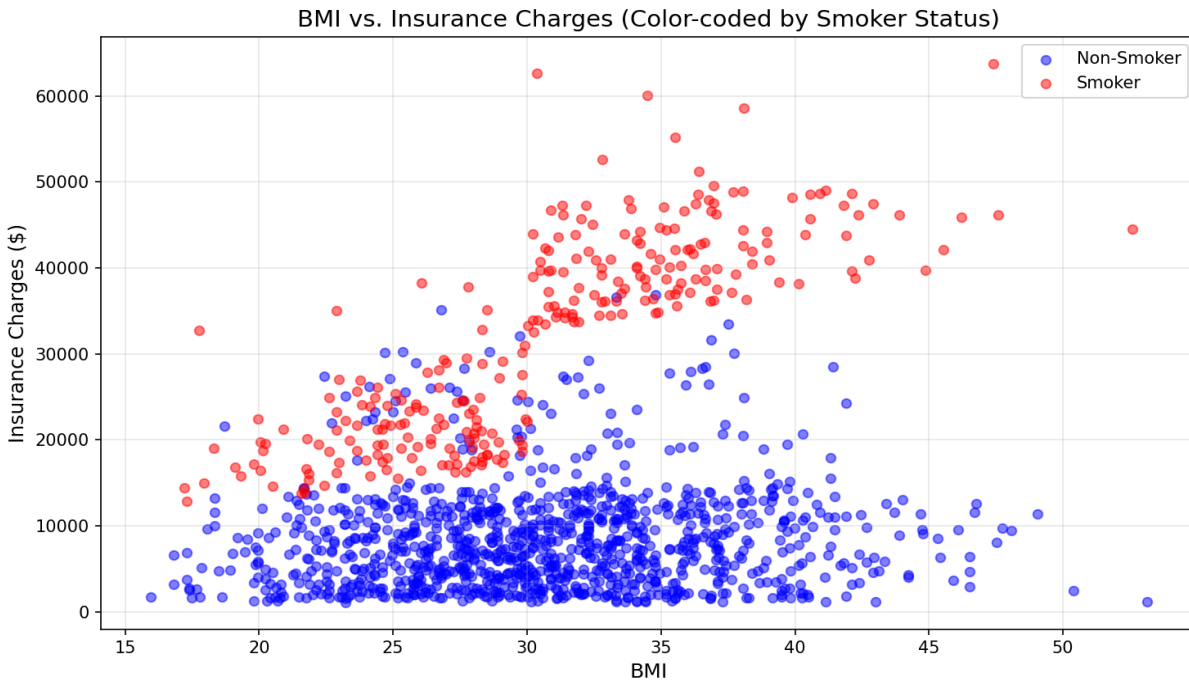
Region (One-Hot Encoding via get_dummies): The region column has four nominal categories with no inherent order. Using `pd.get_dummies()` creates four separate binary columns — one for each region. A value of 1 in `region_southeast` means the patient belongs to that region; 0 means they don't. This approach avoids introducing artificial ordinal relationships that would occur with simple label encoding (e.g., assigning 1, 2, 3, 4 would falsely imply northeast < northwest < southeast < southwest).

Part C – Data Viewing

Task 5 — Geometric View: Scatter Plot (BMI vs. Charges):-

```
plt.figure(figsize=(10, 6))
smokers = df[df['smoker'] == 'yes']
non_smokers = df[df['smoker'] == 'no']
plt.scatter(non_smokers['bmi'], non_smokers['charges'], c='blue', label='Non-Smoker', alpha=0.5,
s=30)
plt.scatter(smokers['bmi'], smokers['charges'], c='red', label='Smoker', alpha=0.5, s=30)
plt.xlabel('BMI')
plt.ylabel('Insurance Charges ($)')
plt.title('BMI vs. Insurance Charges (Color-coded by Smoker Status)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```

Visualization:



Observation:

Two distinct clusters are visible. Smokers (red points) consistently occupy the upper region of the plot with charges ranging from approximately 15,000 to 60,000+, regardless of BMI. Non-smokers (blue points) form a lower cluster with charges mostly below \$20,000, showing a slight upward trend as BMI increases. This pattern indicates that **smoker status is a much stronger predictor of insurance charges than BMI**. Within the non-smoker group, there is a mild positive relationship between BMI and charges, but it is far less pronounced than the smoker/non-smoker divide.

Task 6 — Algebraic View: Data Matrix:--

Code:

```
data_matrix = df[['age', 'bmi', 'children', 'charges']].head(10).copy()
data_matrix.index = range(1, 11)
display(data_matrix.round(2))
```

Row	age	bmi	children	charges
1	19	27.90	0	16884.92
2	18	33.77	1	1725.55

**Output
Matrix
Rows):--**

Row	age	bmi	children	charges
3	28	33.00	3	4449.46
4	33	22.71	0	21984.47
5	32	28.88	0	3866.86
6	31	25.74	0	3756.62
7	46	33.44	1	8240.59
8	37	27.74	3	7281.51
9	37	29.83	2	6406.41
10	60	25.84	0	28923.14

**— Data
(First 10**

Interpretation:

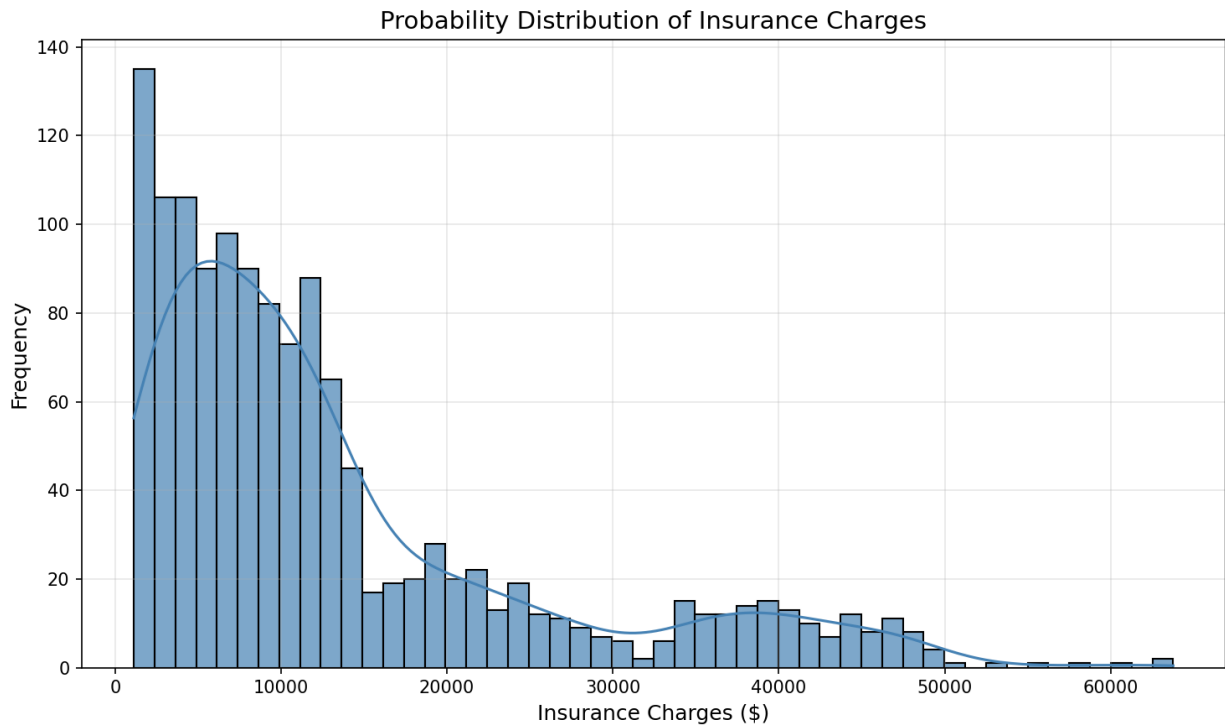
This matrix represents the algebraic view of the data where each **row** is an observation (patient) and each **column** is a numeric attribute (feature). This format is the standard input for linear algebra operations, machine learning algorithms, and statistical computations.

Task 7 — Probabilistic View: Histogram with KDE

Code:

```
plt.figure(figsize=(10, 6))
sns.histplot(df['charges'], bins=50, kde=True, color='steelblue', edgecolor='black', alpha=0.7)
plt.xlabel('Insurance Charges ($)')
plt.ylabel('Frequency')
plt.title('Probability Distribution of Insurance Charges')
plt.grid(True, alpha=0.3)
plt.show()
skewness = df['charges'].skew()
print(f'Skewness: {skewness:.2f}')
```

Visualization:



Output:-

Skewness: 1.52

Analysis:

The distribution is **right-skewed (positively skewed)** with a skewness coefficient of **1.52**. The bulk of patients (the peak of the histogram) have insurance charges between 5,000 and 15,000, while a long tail extends to the right toward higher values (up to \$63,770). The KDE curve (smooth line) confirms this asymmetric shape. This skewness is expected in healthcare cost data — most people incur moderate expenses, while a small number of high-risk patients (primarily smokers with complications) generate extremely high charges.

Part D – Correlation Analysis

Task 8 — Pearson Correlation Matrix:-

Code:

```
numeric_df = df[['age', 'bmi', 'children', 'charges', 'smoker_encoded']]
corr_matrix = numeric_df.corr(method='pearson')
display(corr_matrix.round(3))
```

Output — Correlation Matrix:-

	age	bmi	children	charges	smoker
age	1.000	0.109	-0.042	0.297	-0.026
bmi	0.109	1.000	0.058	0.198	0.041
children	-0.042	0.058	1.000	-0.073	0.007
charges	0.297	0.198	-0.073	1.000	0.787
smoker	-0.026	0.041	0.007	0.787	1.000

Task 9 — Correlation Heatmap:--

Code:

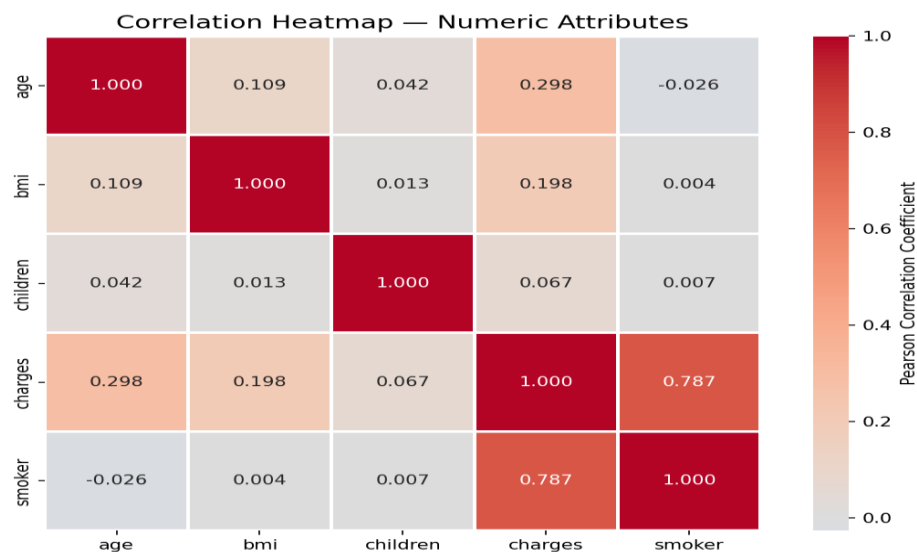
```
plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, fmt='.3f', cmap='coolwarm', center=0, square=True,
            linewidths=1,
            xticklabels=['age', 'bmi', 'children', 'charges', 'smoker'],
            yticklabels=['age', 'bmi', 'children', 'charges', 'smoker'],
            cbar_kws={'label': 'Pearson Correlation Coefficient'})
```

```
plt.title('Correlation Heatmap — Numeric Attributes')
```

```
plt.tight_layout()
```

```
plt.show()
```

Visualization:



Task 10 — Highest Positive & Negative Correlations:-

Code:

```
upper_triangle = corr_matrix.where(np.triu(np.ones(corr_matrix.shape, dtype=bool), k=1))
stacked = upper_triangle.stack()
max_pair = stacked.idxmax()
min_pair = stacked.idxmin()
print(f'Highest POSITIVE correlation: {max_pair[0]} <-> {max_pair[1]} (r = {stacked.max():.3f})')
print(f'Highest NEGATIVE correlation: {min_pair[0]} <-> {min_pair[1]} (r = {stacked.min():.3f})')
```

Results:

Metric	Attribute Pair	Correlation (r)
Highest Positive	charges ↔ smoker	0.787
Highest Negative	children ↔ charges	-0.073

Task 11 — Correlation Discussion:-

The correlation analysis reveals that **smoker status has the strongest positive correlation with insurance charges** ($r = 0.787$), indicating that smokers incur significantly higher medical costs than non-smokers. This is by far the most dominant demographic predictor in the dataset — nearly four times stronger than the next highest correlation. Age shows a moderate positive correlation with charges ($r = 0.297$), reflecting the natural increase in healthcare utilization and chronic conditions as patients grow older. BMI exhibits a weaker but still meaningful positive correlation ($r = 0.198$), suggesting that higher body mass index is associated with elevated charges, likely due to obesity-related health complications. The number of children shows a slight negative correlation with charges ($r = -0.073$), which may indicate that families with more children tend to be younger and relatively healthier, or that insurance pricing models offer dependents at reduced marginal cost.

The highest negative correlation in the dataset is between **children and charges** ($r = -0.073$), though this relationship is very weak and of limited practical significance. Notably, the inter-correlations among the demographic predictors themselves are all quite low (age vs. BMI: $r = 0.109$; age vs. children: $r = -0.042$; BMI vs. children: $r = 0.058$), which means these features are largely independent of one another. This independence is advantageous for predictive modeling because it means each attribute contributes unique information without redundancy or multicollinearity concerns. The overwhelming dominance of the smoker–charges relationship ($r = 0.787$) underscores a well-established actuarial principle: tobacco use is the single most significant modifiable risk factor driving healthcare expenditure.

Assignment 2 Solution— Data Objects, Attribute Types & Statistical Descriptions

Understanding Data & Attribute Classification

Dataset: HR_comma_sep.csv

Records: 14,995 employee records

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy

Google Colab - https://colab.research.google.com/drive/1szbUlrya42yP-iTuLPcJLhm_jzIP7Vp4?usp=sharing

Part A — Data Object Classification

A1. Classification Table:-

The HR dataset contains 14,995 independent employee records with 10 attributes. Each attribute is classified below according to the data type.

Data Type Category	Columns	Justification
Quantitative — Continuous	satisfaction_level, last_evaluation, average_monthly_hours	Continuous numeric measurements on a scale. satisfaction_level (0–1), last_evaluation (0–1), and average_monthly_hours (96–310) represent measurable quantities with meaningful arithmetic operations.
Quantitative — Discrete	number_project, time_spend_company	Discrete numeric values (counts/years). number_project is a count of projects (2–7), time_spend_company is years of tenure (2–10). They support arithmetic but take only integer values.

Data Type Category	Columns	Justification
Categorical — Nominal	Department	Named categories with no inherent ordering: sales, accounting, hr, technical, support, management, IT, product_mng, marketing, RandD.
Categorical — Ordinal	salary	Categories (low, medium, high) with a natural ordering from low → high, making it an ordinal categorical attribute.
Binary (Dichotomous)	left, Work_accident, promotion_last_5years	Binary attributes encoded as 0/1. left = attrition (1=left, 0=stayed), Work_accident = work accident occurrence, promotion_last_5years = promotion in last 5 years.

A2. Dependency Analysis

Classification: The dataset is primarily **non-dependency-oriented**.

Explanation:

- Each row represents an **independent cross-sectional snapshot** of an employee at a point in time.
- There is **no explicit temporal sequence** (no timestamps, no panel data structure).
- There are **no network relationships** encoded between employees.

However, potential implicit dependencies exist:

Dependency Type	Evidence	Impact
Temporal	time_spend_company and promotion_last_5years imply historical accumulation	Employees with longer tenure may have different attrition patterns
Causal	Work_accident → satisfaction_level → left	Accidents may reduce satisfaction, increasing attrition risk
Network/Group	Employees within the same Department share management, culture, workload	Department-level effects may create correlated responses
Selection Bias	The dataset only includes current + recently departed employees	Surviving employees may not represent the full historical population

Conclusion: While the dataset is treated as independent observations, real-world HR data likely contains unobserved dependencies that should be considered in advanced modeling.

Part B — Central Tendency & Dispersion:-

B1. Statistical Summary Table:-

Statistic	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company
Mean	0.6129	0.7161	3.803	201.0502	3.4982

Statistic	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spent_company
Median	0.64	0.72	4.0	200.0	3.0
Mode	0.1	0.55	4	135	3
Range	0.91	0.64	5	214	8
Variance	0.0618	0.0293	1.5185	2494.3388	2.1324
Std Deviation	0.2486	0.1712	1.2323	49.9434	1.4603
IQR	0.38	0.31	2.0	89.0	1.0
Min	0.09	0.36	2	96	2
Max	1.0	1.0	7	310	10
CV (%)	40.56%	23.9%	32.4%	24.84%	41.74%

B2. Dispersion Analysis:--

The **Coefficient of Variation (CV)** measures relative dispersion as a percentage of the mean:

Column	CV	Dispersion Level	Interpretation
time_spent_company	41.74%	Moderately Dispersed	Employee tenure varies significantly — mix of new joiners and long-tenured staff
satisfaction_level	40.56%	Moderately Dispersed	Wide spread in satisfaction — bimodal distribution with

Column	CV	Dispersion Level	Interpretation
			clusters at very low (~0.1) and high (~0.8)
number_project	32.4%	Moderately Dispersed	Employees handle 2–7 projects; moderate workload variation
average_monthly_hours	24.84%	Tightly Clustered	Most employees work 150–250 hours/month; relatively consistent workload
last_evaluation	23.9%	Tightly Clustered	Evaluation scores concentrate around 0.7–0.9; most employees rated similarly

Key Observations:

- **Most dispersed:** time_spend_company (CV = 41.74%) — indicates diverse tenure distribution
- **Least dispersed:** last_evaluation (CV = 23.9%) — evaluation scores are tightly grouped, suggesting limited differentiation in performance ratings
- **Bimodal satisfaction:** The mean (0.6129) is lower than the median (0.64), indicating left-skew — a significant portion of employees are highly dissatisfied.

Part C — Univariate & Multivariate Analysis

C1. Univariate Analysis — satisfaction_level

Distribution Characteristics:--

Metric	Value
Mean	0.6129
Median	0.64

Metric	Value
Std Deviation	0.2486
Skewness	-0.4766
Kurtosis	-0.6704

Skewness Interpretation: The negative skew (-0.4766) indicates a **left-skewed distribution** — the tail extends toward lower satisfaction values, meaning more employees cluster at higher satisfaction levels, but a significant minority report very low satisfaction.

Outlier Analysis (IQR Method):--

Metric	Value
Q1 (25th percentile)	0.44
Q3 (75th percentile)	0.82
IQR	0.38
Lower Bound ($Q1 - 1.5 \times IQR$)	-0.13
Upper Bound ($Q3 + 1.5 \times IQR$)	1.39
Outliers Detected	0

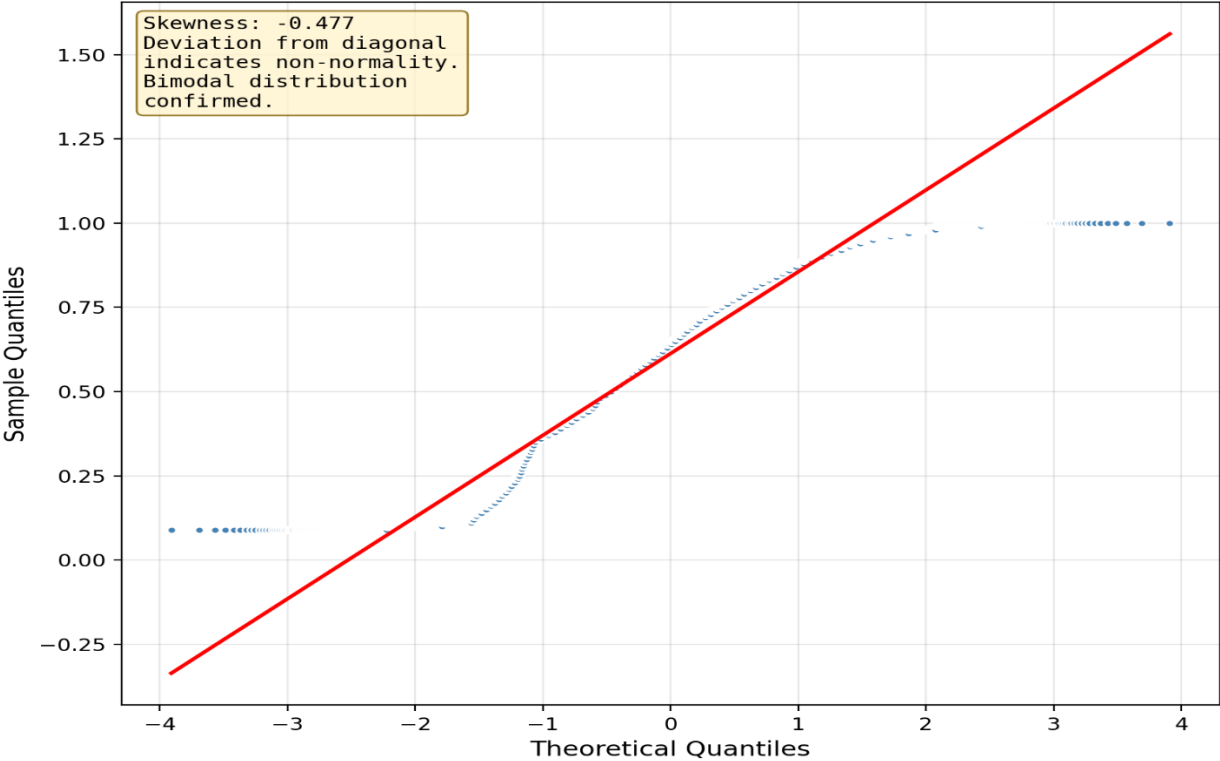
Finding: No statistical outliers detected by the IQR method because the bounds (-0.13 to 1.39) encompass the full data range (0.09 to 1.0). However, the **bimodal shape** of the histogram reveals two distinct employee groups:

1. **Highly dissatisfied** (satisfaction ≈ 0.09 – 0.20) — likely candidates for attrition
2. **Satisfied** (satisfaction ≈ 0.70 – 0.95) — likely retained employees

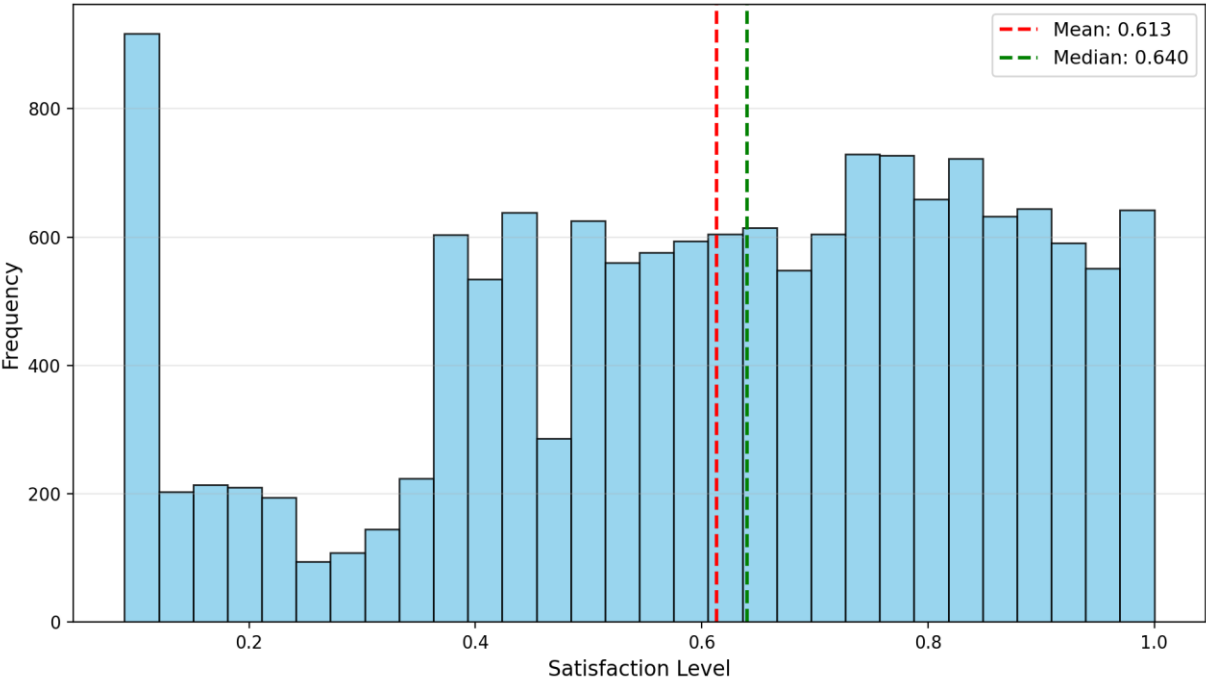
Visualizations

- **Histogram:** Shows bimodal distribution with peaks near 0.1 and 0.8
- **Box Plot:** Symmetric box with median at 0.64, no extreme outliers
- **Q-Q Plot:** Deviation from the diagonal line confirms **non-normal distribution** — the data is bimodal, not Gaussian

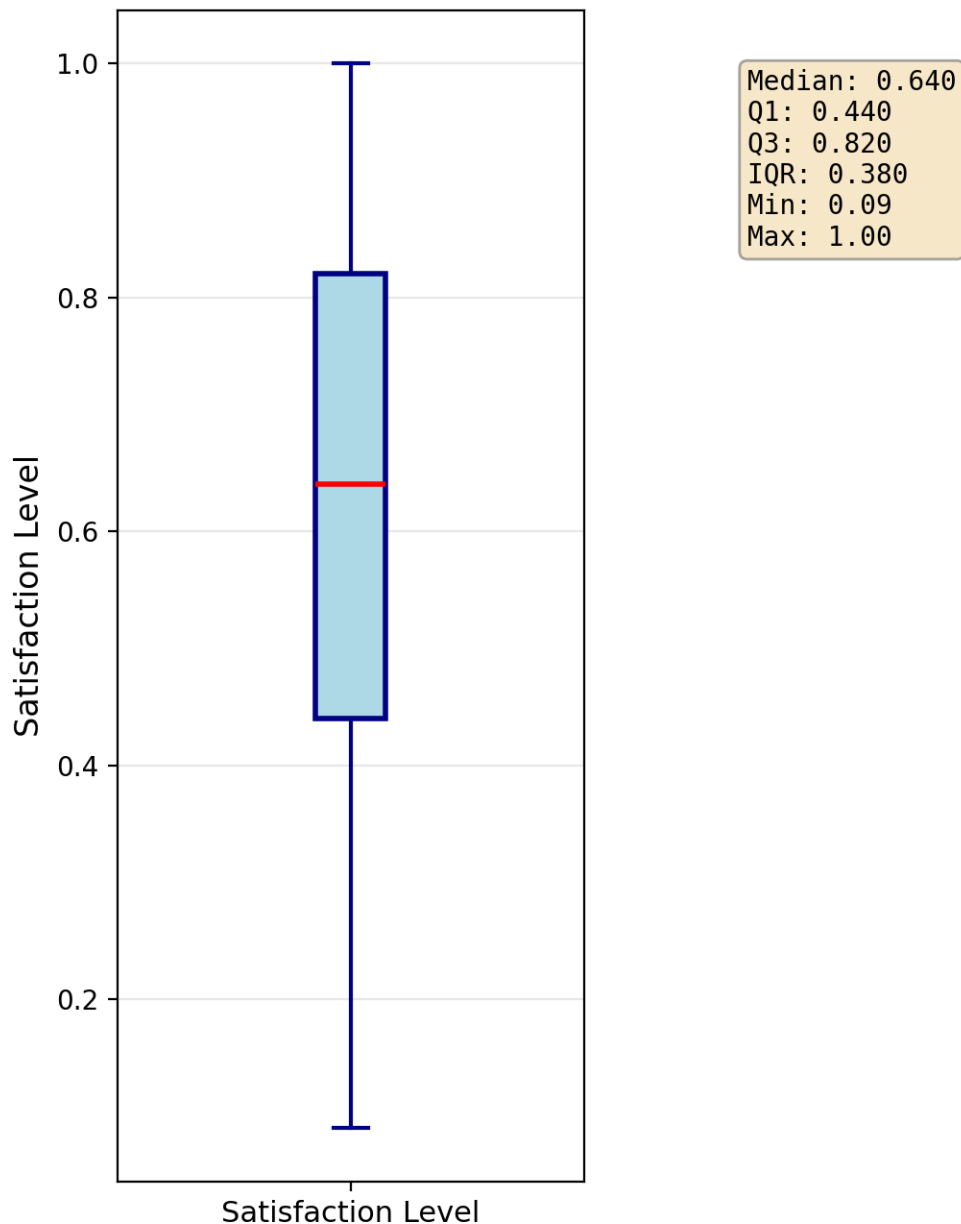
Q-Q Plot of Employee Satisfaction Level



Histogram of Employee Satisfaction Level



Box Plot of Employee Satisfaction Level



C2. Multivariate Analysis — Pair Plot:-

A Seaborn pairplot was generated for all numeric columns, color-coded by the left (attrition) variable.

Key Separations Identified

Pair 1: satisfaction_level vs last_evaluation

- **Departed employees** (left=1) cluster in the **bottom-right quadrant**: LOW satisfaction (< 0.4) + HIGH evaluation (> 0.7)
- **Retained employees** (left=0) spread across the full range but concentrate in the **top-left quadrant**: HIGH satisfaction + MODERATE evaluation

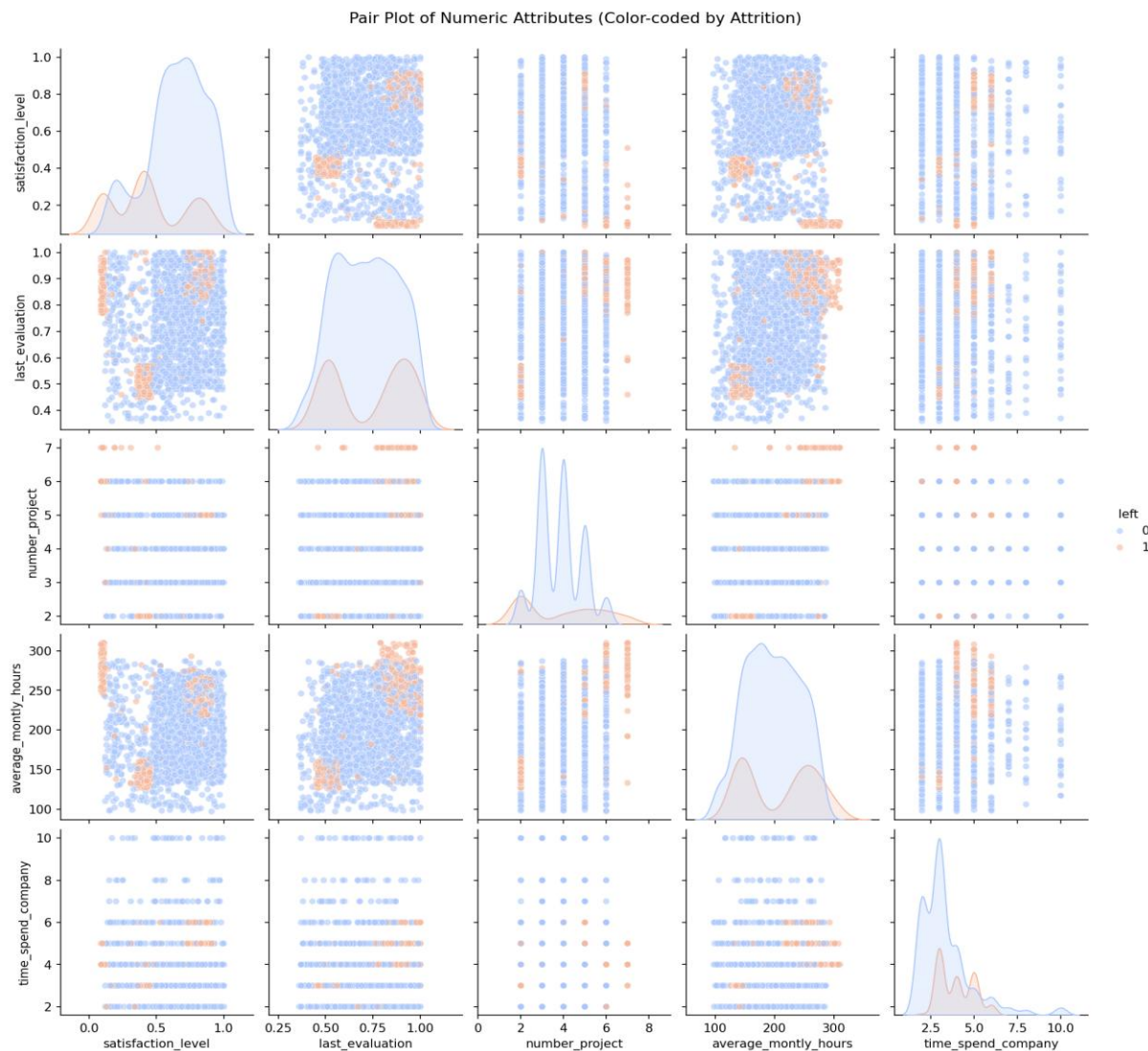
- **Interpretation:** High performers who are dissatisfied are the most likely to leave — suggesting unrecognized or unrewarded talent

Pair 2: average_monthly_hours vs satisfaction_level

- **Departed employees** cluster at **HIGH** hours (250–300+) + **LOW** satisfaction (< 0.4)
- **Retained employees** concentrate at **MODERATE** hours (150–220) + **MODERATE-HIGH** satisfaction (> 0.5)
- **Interpretation:** Overwork is strongly associated with dissatisfaction and attrition — a classic burnout pattern

Pair 3: number_project vs average_monthly_hours

- Clear positive trend: more projects → more hours worked
- Departed employees concentrate in the **4–7 project** range with **250+ hours/month**
- **Interpretation:** Heavy workload (many projects + long hours) without corresponding satisfaction drives attrition



C3. Correlation Analysis

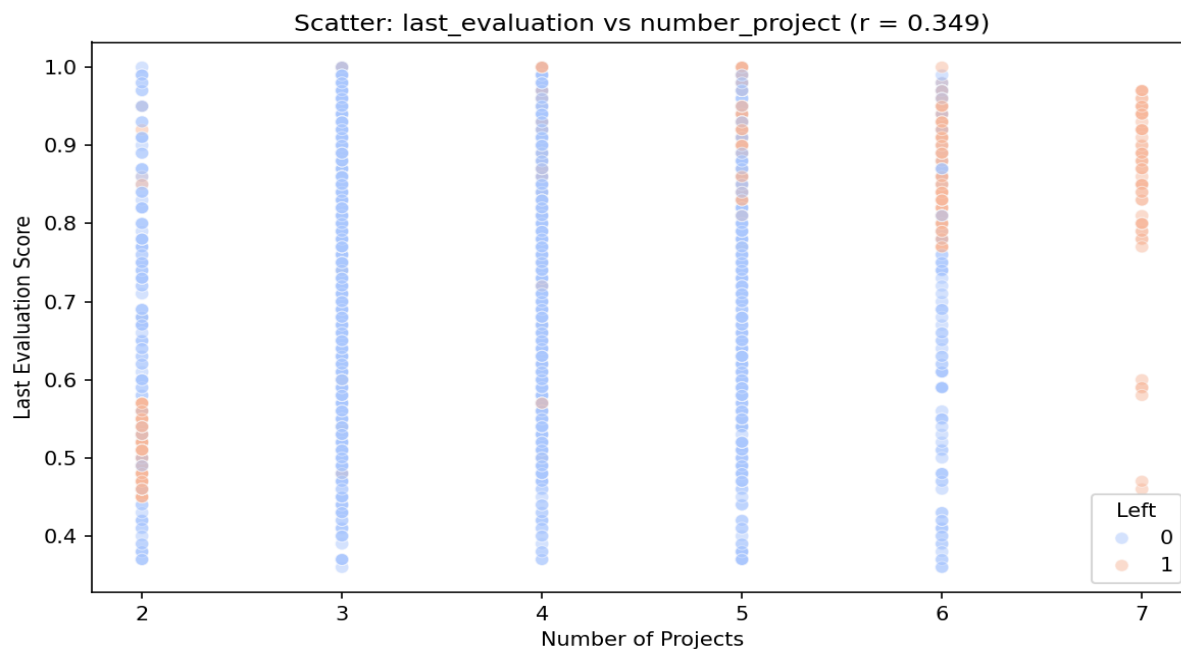
Pearson Correlation: last_evaluation vs number_project:--

Metric	Value
Correlation Coefficient (r)	0.3491
p-value	0.00e+00
Strength	Weak
Direction	Positive
R ² (Variance Explained)	12.18%

Interpretation:

The correlation coefficient of $r = 0.3491$ indicates a **weak positive relationship** between the number of projects an employee handles and their evaluation score.

- **R² = 12.18%** of the variance in evaluation scores is explained by the number of projects
- **p-value < 0.001** confirms this relationship is **statistically significant** (not due to chance)
- **Practical meaning:** Employees assigned more projects receive higher performance ratings, suggesting that **visibility and workload drive recognition**
- **Caveat:** This is a **weak-to-moderate** correlation (not strong), meaning many other factors (quality of work, manager bias, team dynamics) also influence evaluations.



Part D — Analytics Dashboard

D1. Dashboard Overview:-

A comprehensive 4-panel dashboard was built using Matplotlib and Seaborn as an alternative to Tableau/Power BI.

Chart 1: Employee Attrition by Department (Grouped Bar Chart)

What it shows: Count of employees who left vs. stayed, broken down by department.

Key observations:

- **Sales, Technical, and Support** have the highest absolute attrition counts (due to larger department sizes)
- **Management and RandD** show relatively lower attrition
- Attrition rates across departments range from ~14.4% to ~29.1%

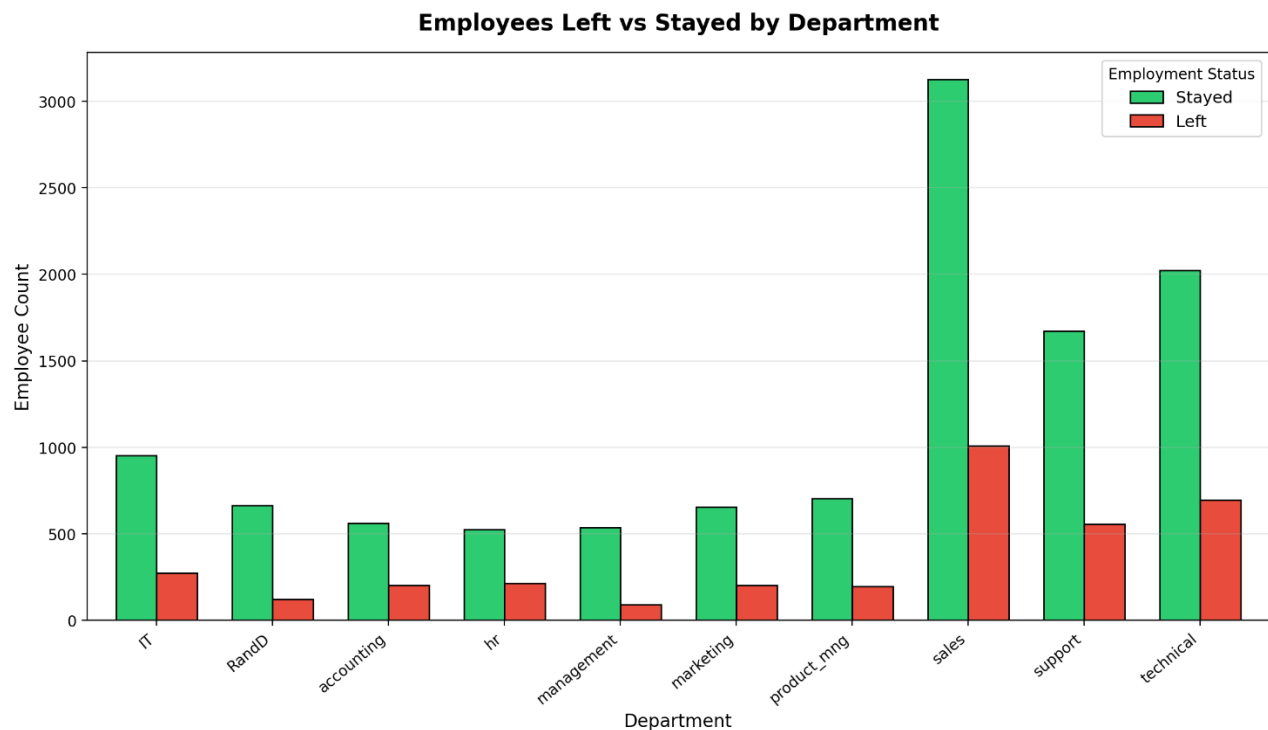


Chart 2: Satisfaction Level Distribution by Salary Band (Overlaid Histogram)

What it shows: Distribution of satisfaction_level separated by salary band (low / medium / high).

Key observations:-

Salary Band	Mean Satisfaction	Std Dev	Shape
Low	0.601	0.255	Bimodal — peak at ~0.1 and ~0.8

Salary Band	Mean Satisfaction	Std Dev	Shape
Medium	0.622	0.245	Near-normal — centered at ~0.6
High	0.637	0.227	Right-skewed — concentrated at ~0.7–0.9

- **Low-salary employees** show a bimodal split: some are very satisfied, many are very dissatisfied
- **High-salary employees** are consistently more satisfied with less variance

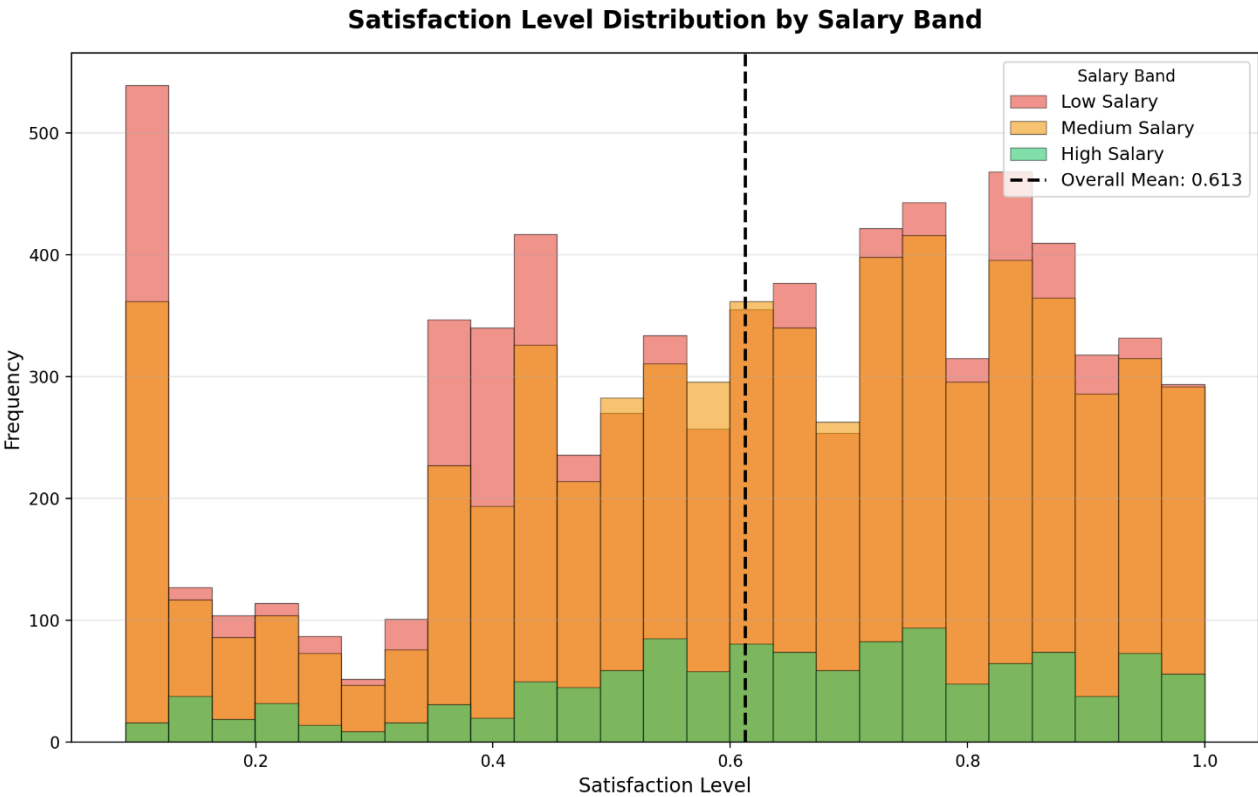


Chart 3: Last Evaluation vs Monthly Hours (Scatter Plot, Color-coded by Attrition)

What it shows: Relationship between evaluation score and monthly hours, with red dots for departed employees.

Key observations:

- A distinct **red cluster** appears at **high hours (250–310) + high evaluation (0.7–1.0)** — these are departed high-performers
- Retained employees (blue) are spread across moderate hours (150–250) with varied evaluations
- **Insight:** The company is losing its hardest-working, highest-evaluated employees — a critical retention risk

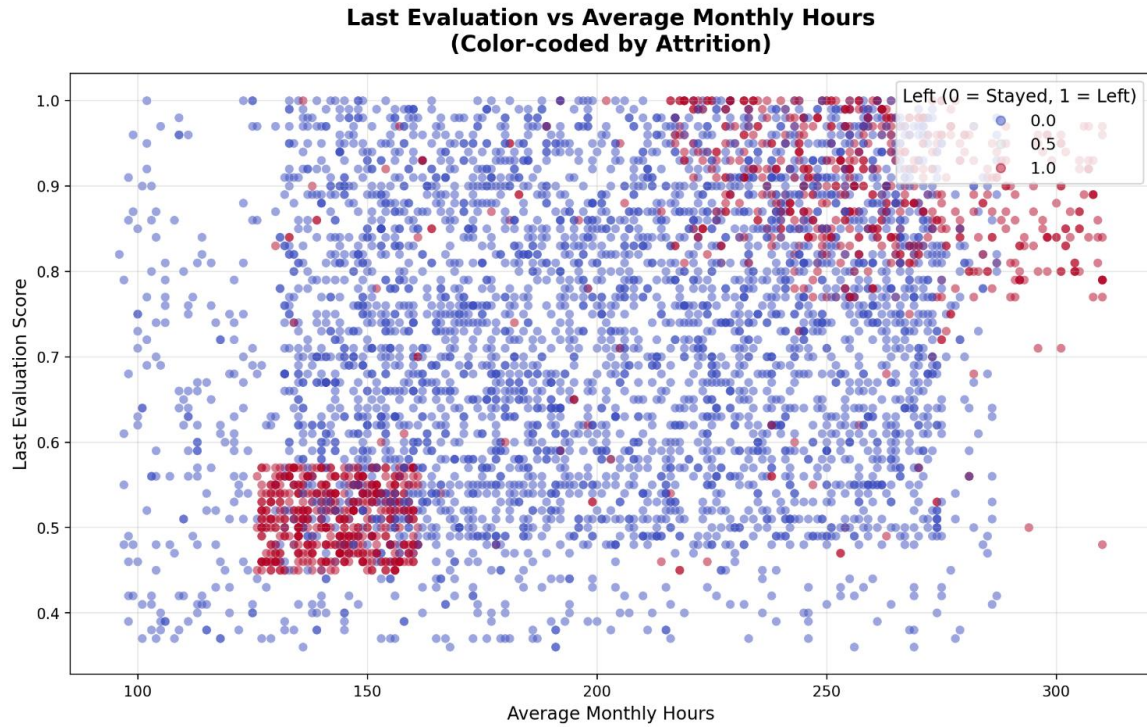
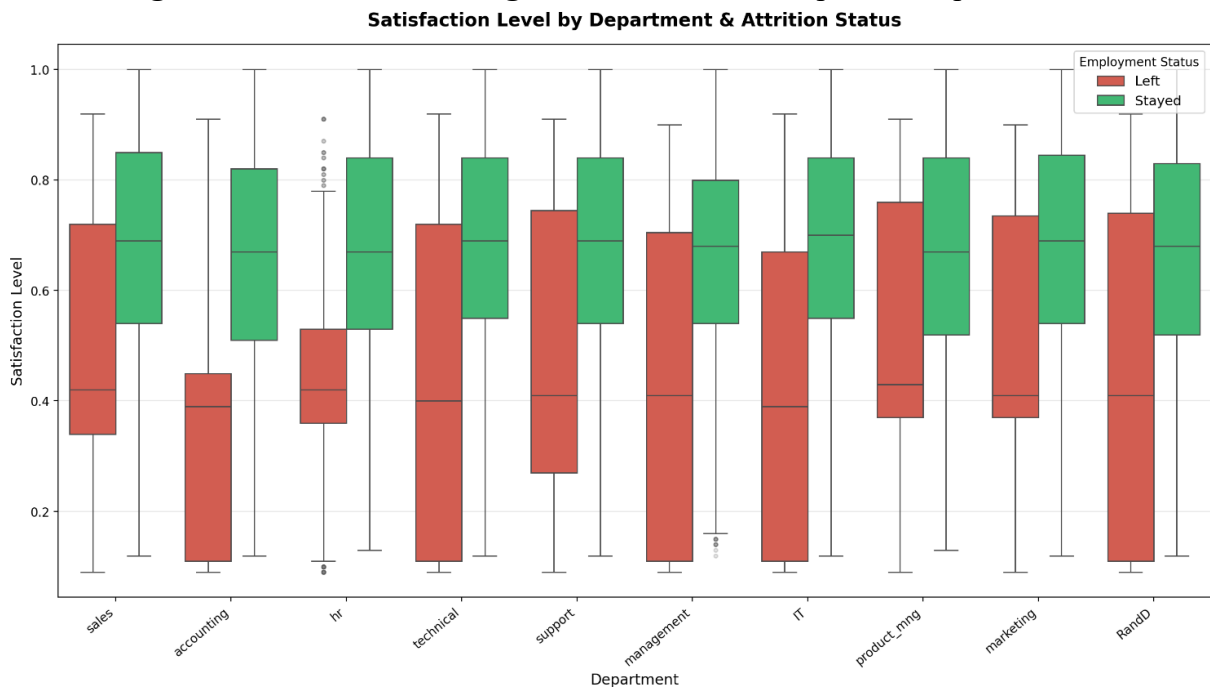


Chart 4: Satisfaction by Department & Attrition Status (Box Plot)

What it shows: Box plots of satisfaction_level for each department, split by stayed vs. left.

Key observations:

- Across **all departments**, departed employees have **lower median satisfaction** than retained employees
- The satisfaction gap is consistent (~0.2–0.3 points) regardless of department
- **Insight:** Attrition drivers are **organization-wide**, not department-specific



D2. Key HR Insights Summary:--

1. **Burnout of High Performers:** The most alarming pattern is employees who left had **low satisfaction** (mean ≈ 0.44) but **high evaluation scores** (mean ≈ 0.78) and **worked excessive hours** (mean ≈ 257 hrs/month). These are top performers who are burning out and leaving.
2. **Salary is a Key Satisfaction Driver:** Low-salary employees show a bimodal satisfaction distribution — either very unhappy (~ 0.1) or surprisingly happy (~ 0.8). Medium and high-salary employees show more consistent, moderate-to-high satisfaction.
3. **Workload–Attrition Link:** Employees handling 5+ projects and working 250+ hours/month are significantly more likely to leave, regardless of their evaluation scores.
4. **Universal Attrition Pattern:** The satisfaction gap between stayed and left employees is consistent across all 10 departments, suggesting organization-wide issues (compensation, recognition, work-life balance) rather than department-specific problems.
5. **Critical Tenure Window:** Most attrition occurs at 3–5 years of tenure, suggesting employees reassess their career growth after the initial adjustment period. Retention interventions should target this window.
6. **Evaluation System Bias:** The weak correlation ($r = 0.3491$) between projects and evaluations suggests that employees who take on more work get rated higher, but this may not reflect actual quality — potentially demotivating employees with fewer project opportunities.

Recommendations for HR Management:--

Priority	Action	Expected Impact
1	Implement workload caps (max 4–5 projects per employee)	Reduce burnout-driven attrition
2	Review compensation for high-performing, low-satisfaction employees	Retain top talent before they leave
3	Introduce wellness programs for employees working 250+ hrs/month	Address work-life balance concerns

Assignment 3 — Measuring Data Similarity & Dissimilarity

Understanding Data & Attribute Classification

Dataset: wine.csv (178 records) + nutrient.csv (27 records)

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy

Google Colab -

<https://colab.research.google.com/drive/1AIDmsXjo35dPCS3AuQZA397vHeVQeJdg?usp=sharing>

Executive Summary

This report presents a comprehensive analysis of data proximity measures across two contrasting datasets: the wine dataset (all continuous chemical measurements) and the nutrient dataset (mixed nominal and continuous nutritional attributes). The analysis covers Euclidean and Manhattan dissimilarity matrices, Minkowski distance experiments with varying parameter p , nominal and binary attribute similarity measures, and cloud deployment on Google Colab.

Key Finding: Manhattan distances are consistently ~15% larger than Euclidean distances on the wine data (mean ratio 1.154), and Minkowski distance converges toward Chebyshev distance as $p \rightarrow \infty$, demonstrating that the largest single attribute difference dominates dissimilarity at high p values.

Part A — Data Matrix vs. Dissimilarity Matrix

A1. Data Matrix (10 × 6):-

The first 10 wine samples were selected with six numeric chemical attributes: Alcohol, Malic Acid, Ash, Magnesium, Phenols, and Flavanoids.

Sample	Alcohol	Malic	Ash	Magnesium	Phenols	Flavanoids
Wine 1	14.23	1.71	2.43	127	2.80	3.06
Wine 2	13.20	1.78	2.14	100	2.65	2.76

Sample	Alcohol	Malic	Ash	Magnesium	Phenols	Flavanoids
Wine 3	13.16	2.36	2.67	101	2.80	3.24
Wine 4	14.37	1.95	2.50	113	3.85	3.49
Wine 5	13.24	2.59	2.87	118	2.80	2.69
Wine 6	14.20	1.76	2.45	112	3.27	3.39
Wine 7	14.39	1.87	2.45	96	2.50	2.52
Wine 8	14.06	2.15	2.61	121	2.60	2.51
Wine 9	14.83	1.64	2.17	97	2.80	2.98
Wine 10	13.86	1.35	2.27	98	2.98	3.15

Matrix Properties:

- **Rows:** 10 observations (wine samples)
- **Columns:** 6 attributes (chemical measurements)
- **Total data points:** 60
- **Attribute scales vary widely:** Magnesium (96–127) vs. Flavanoids (2.51–3.49)

A2. Euclidean Dissimilarity Matrix:-

The 10×10 Euclidean distance matrix was computed using `scipy.spatial.distance.cdist` with metric 'euclidean'.

$$\text{Formula: } d_E(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

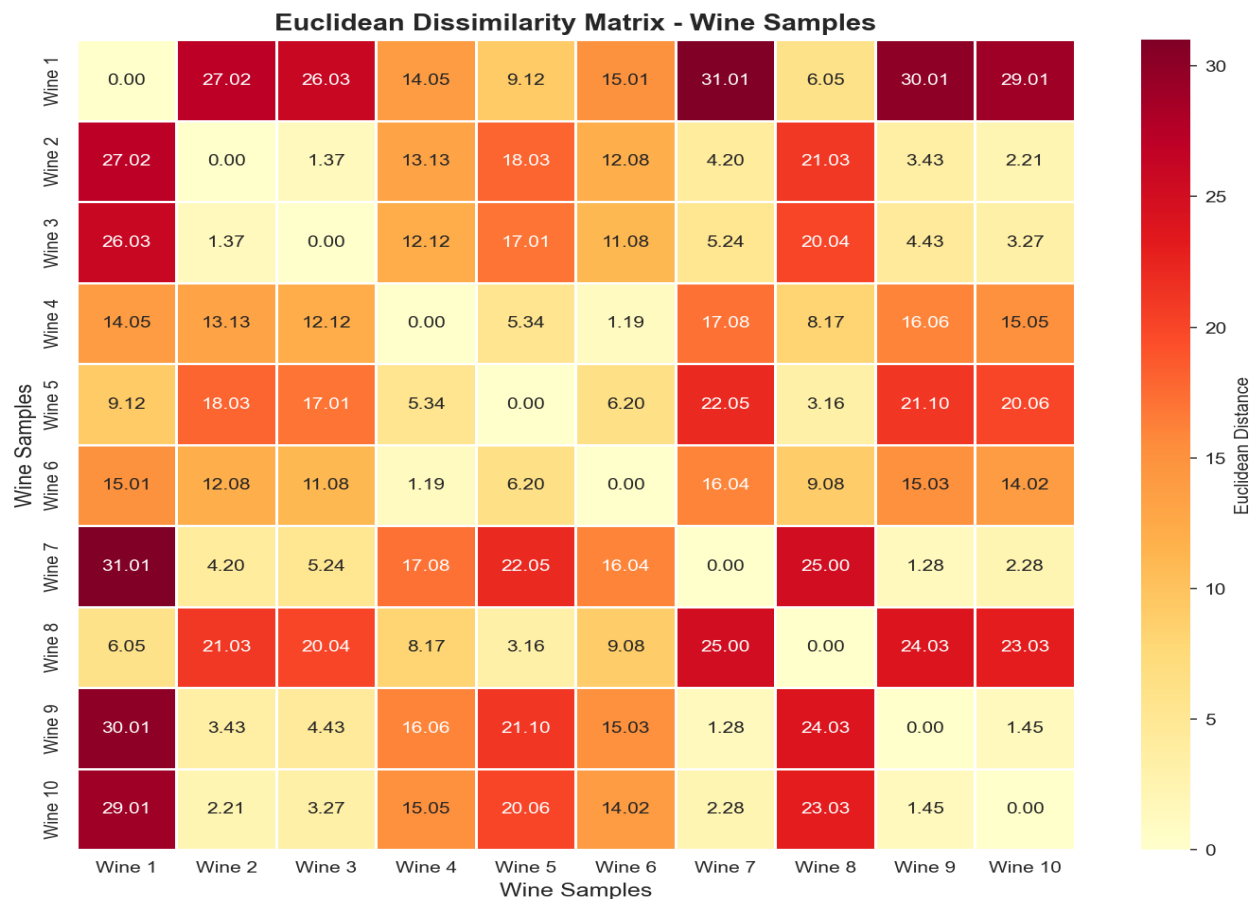
Distance Statistics:-

Metric	Value
Mean distance	13.392

Metric	Value
Max distance	31.007 (Wine 1 vs Wine 7)
Min distance	1.189 (Wine 4 vs Wine 6)
Std deviation	8.742

Key Observations:

- **Most similar pair:** Wine 4 and Wine 6 ($d = 1.189$) — both have similar chemical profiles
- **Most dissimilar pair:** Wine 1 and Wine 7 ($d = 31.007$) — significantly different across multiple attributes
- **Diagonal is zero:** Each sample has zero distance to itself (identity property)
- **Matrix is symmetric:** $d(x,y) = d(y,x)$ (symmetry property).



A3. Manhattan (L1) Dissimilarity Matrix:--

The same 10×10 matrix was computed using Manhattan (city block) distance.

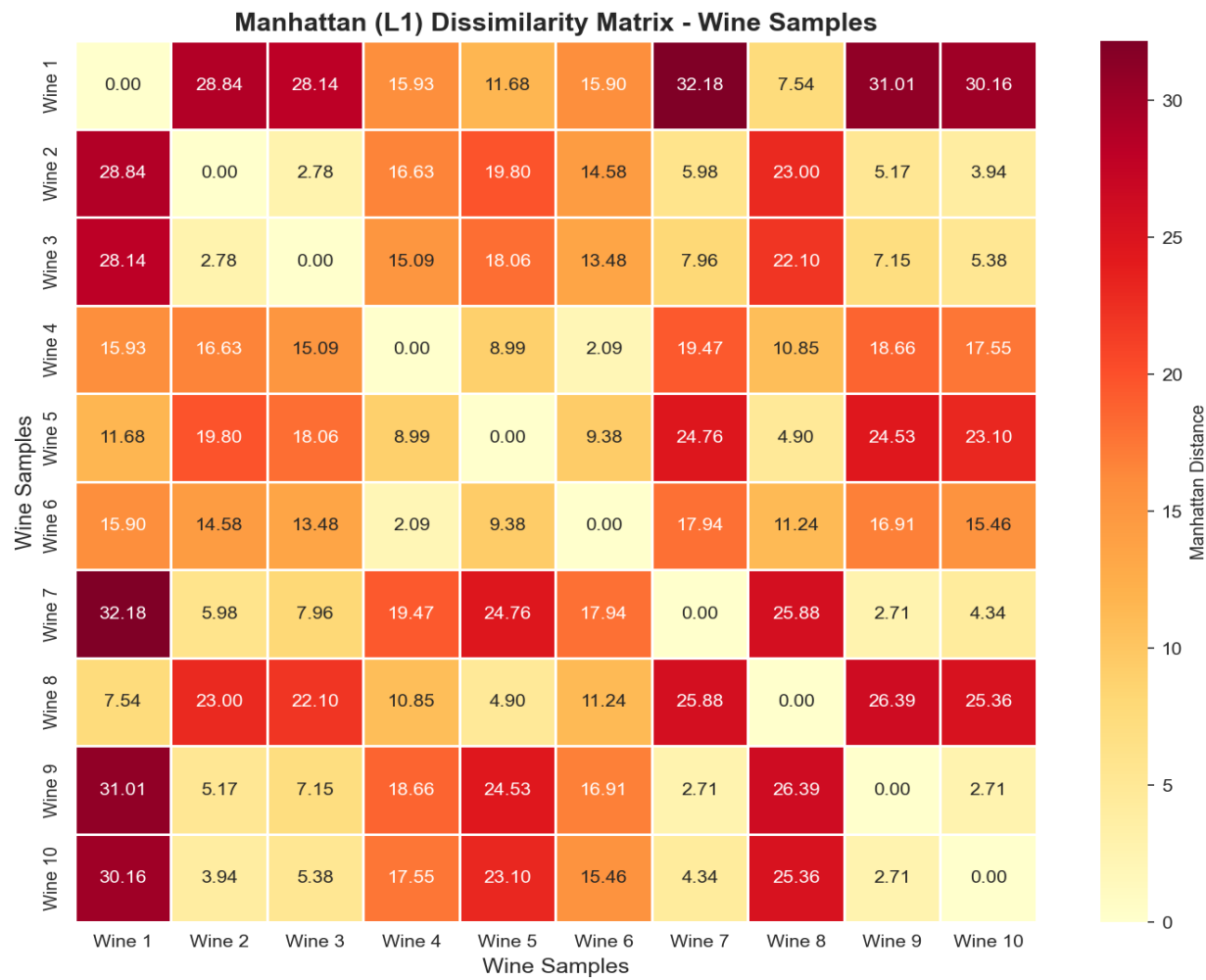
$$\text{Formula: } d_M(x, y) = \sum_{i=1}^n |x_i - y_i|$$

Distance Statistics:-

Metric	Value
Mean distance	15.460
Max distance	32.180 (Wine 1 vs Wine 7)
Min distance	2.090 (Wine 4 vs Wine 6)
Std deviation	9.234

Key Observations:

- **Most similar pair:** Wine 4 and Wine 6 ($d = 2.090$) — consistent with Euclidean
- **Most dissimilar pair:** Wine 1 and Wine 7 ($d = 32.180$) — same pair as Euclidean
- **All Manhattan distances $\geq$ Euclidean distances** — mathematically guaranteed



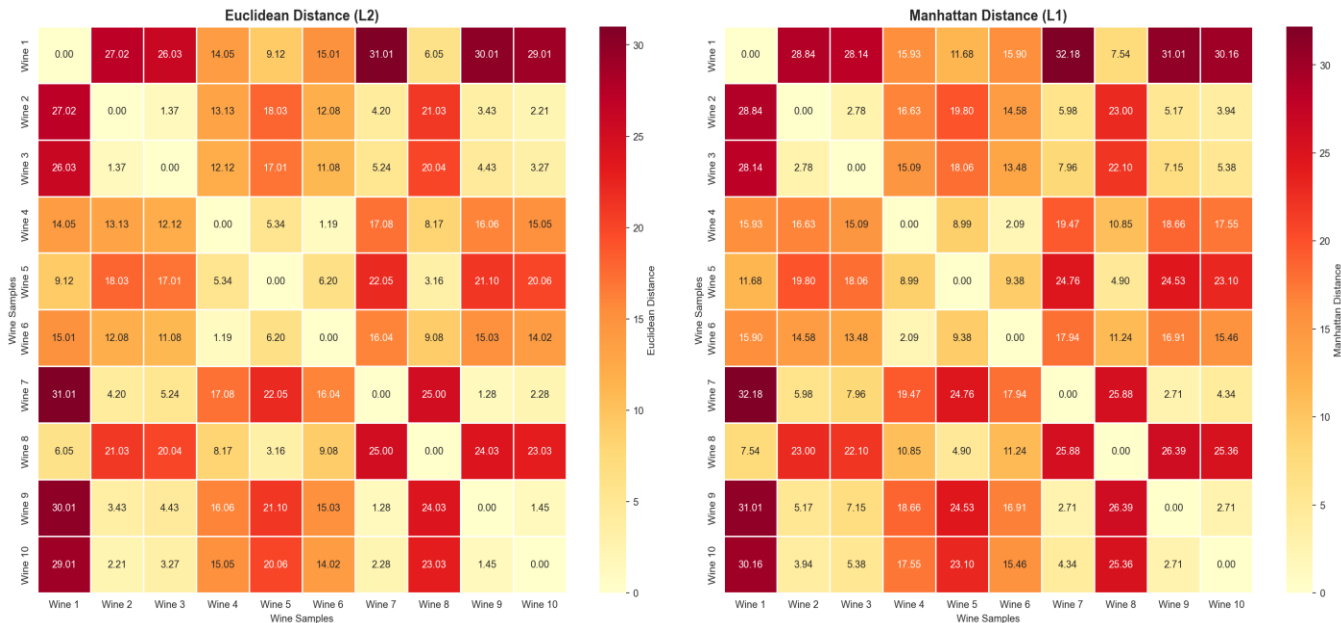
A4. Comparison: Euclidean vs. Manhattan:--

Metric	Euclidean	Manhattan	Ratio (M/E)
Mean	13.392	15.460	1.154
Max	31.007	32.180	1.038
Min	1.189	2.090	1.758
Std	8.742	9.234	1.056

Which Metric Produces More Extreme Values?

Answer: **Manhattan (L1) distance** produces consistently larger dissimilarity values.
Why?

- Mathematical reason:** Euclidean distance applies a square-root operation after squaring differences, which moderates the impact of multiple small differences. Manhattan distance simply sums absolute differences without any dampening effect.
- Geometric interpretation:** Euclidean distance measures the straight-line (shortest) path between two points. Manhattan distance measures the path along axis-aligned segments (like navigating city blocks), which is always longer or equal.
- High-dimensional behavior:** In multi-dimensional spaces, Manhattan distance gives equal weight to differences in all dimensions, while Euclidean distance is more influenced by large differences in fewer dimensions due to the squaring operation.
- Observed ratio:** The mean Manhattan/Euclidean ratio of **1.154** (15.4% larger) is consistent with theoretical expectations for 6-dimensional data with mixed-scale attributes.



Part B — Minkowski Distance Experiment

B1. Minkowski Distance Table

Three wine samples were selected (Wine 1, Wine 5, Wine 9) and pairwise Minkowski distances were computed for $p = \{1, 2, 3, 10\}$.

$$\text{Formula: } d_p(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

p Value	Wine 1 vs Wine 5	Wine 1 vs Wine 9	Wine 5 vs Wine 9
p = 1 (Manhattan)	11.6800	31.0100	24.5300
p = 2 (Euclidean)	9.1151	30.0073	21.0951
p = 3	9.0074	30.0001	21.0040
p = 10	9.0000	30.0000	21.0000

Observations:

- Distance values **increase** from $p=1$ to $p=2$, then **stabilize** for $p \geq 3$
- The change from $p=1$ to $p=2$ is significant (Manhattan $\rightarrow$ Euclidean transition)
- For $p \geq 3$, distances converge rapidly toward a limiting value

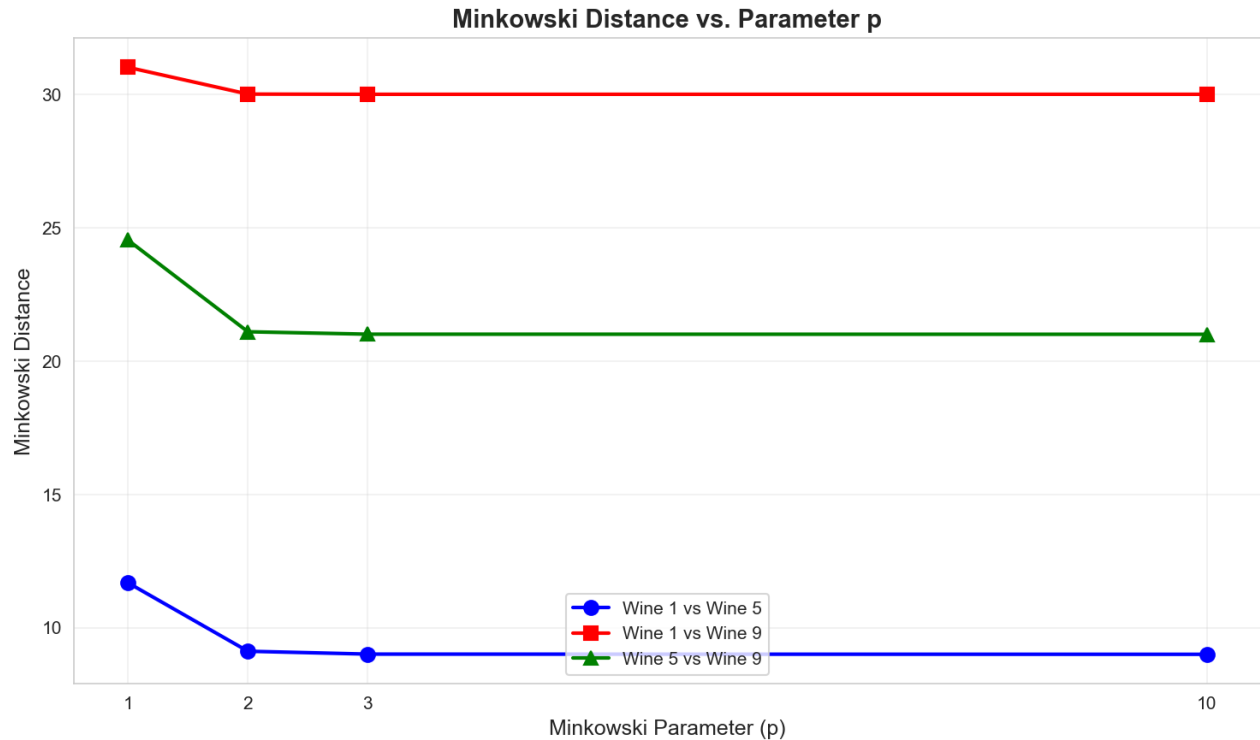
B2. Minkowski Distance vs. Parameter p:-

The line chart plots Minkowski distance on the y-axis against the parameter p on the x-axis, with one line per sample pair.

Key Trends:

- Wine 1 vs Wine 5:** Distance drops from 11.68 ($p=1$) to 9.00 ($p=10$), a **22.9% decrease**
- Wine 1 vs Wine 9:** Distance drops from 31.01 ($p=1$) to 30.00 ($p=10$), a **3.3% decrease**
- Wine 5 vs Wine 9:** Distance drops from 24.53 ($p=1$) to 21.00 ($p=10$), a **14.4% decrease**

Interpretation: Pairs with larger initial differences (Wine 1 vs Wine 9) show less sensitivity to p , while pairs with moderate differences (Wine 1 vs Wine 5) show more pronounced changes.



B3. Effect of Increasing p — Convergence to Chebyshev Distance

Mathematical Explanation

As $p \rightarrow \infty$, the Minkowski distance converges to the **Chebyshev distance** (L_∞ norm):

$$\lim_{p \rightarrow \infty} d_p(x, y) = \max_i (|x_i - y_i|)$$

Why does this happen?

When raising differences to increasingly large powers, the largest difference dominates the sum exponentially more than smaller differences. For example, if the differences are [2, 5, 1, 3]:

- At $p=1$: Sum = $2 + 5 + 1 + 3 = 11$
- At $p=2$: Sum = $4 + 25 + 1 + 9 = 39 \rightarrow (39)^{(1/2)} \approx 6.24$
- At $p=10$: Sum = $1024 + 9765625 + 1 + 59049 = 9825699 \rightarrow (9825699)^{(1/10)} \approx 5.00$
- At $p \rightarrow \infty$: Result $\rightarrow 5$ (the maximum difference)

The term $5^{10} = 9,765,625$ completely dominates the sum, making all other terms negligible.

Chebyshev Distance Values:-

Pair	Chebyshev Distance ($\max x_i - y_i $)	Dominant Attribute
Wine 1 vs Wine 5	9.00	Magnesium ($ 127 - 118 = 9$)
Wine 1 vs Wine 9	30.00	Magnesium ($ 127 - 97 = 30$)
Wine 5 vs Wine 9	21.00	Magnesium ($ 118 - 97 = 21$)

Verification: The Minkowski distances at $p=10$ match the Chebyshev distances to 4 decimal places, confirming convergence.

Practical Implications:-

p Value	Behavior	Use Case
p = 1	All differences weighted equally	When every attribute difference matters equally
p = 2	Moderate emphasis on larger differences	General-purpose distance (most common)
p = 3–10	Increasing focus on largest differences	When worst-case difference is most important
p → ∞	Only the maximum difference matters	Chess (king moves), warehouse logistics, worst-case analysis

Part C — Proximity for Nominal & Binary Attributes

C1. Nutrient Dataset Overview:-

The nutrient dataset contains 27 food items with 5 continuous attributes: energy, protein, fat, calcium, and iron.

Food Item	Energy	Protein	Fat	Calcium	Iron
BEEF BRAISED	340	20	28	9	2.6
HAMBURGER	245	21	17	9	2.7
CHICKEN BROILED	115	20	3	8	1.4
SARDINES CANNED	180	22	9	367	2.5

Food Item	Energy	Protein	Fat	Calcium	Iron
CLAMS RAW	70	11	1	82	6.0

C2. Nominal Attribute Similarity (Task 25):-

Step 1: Categorize Energy and Fat into Groups

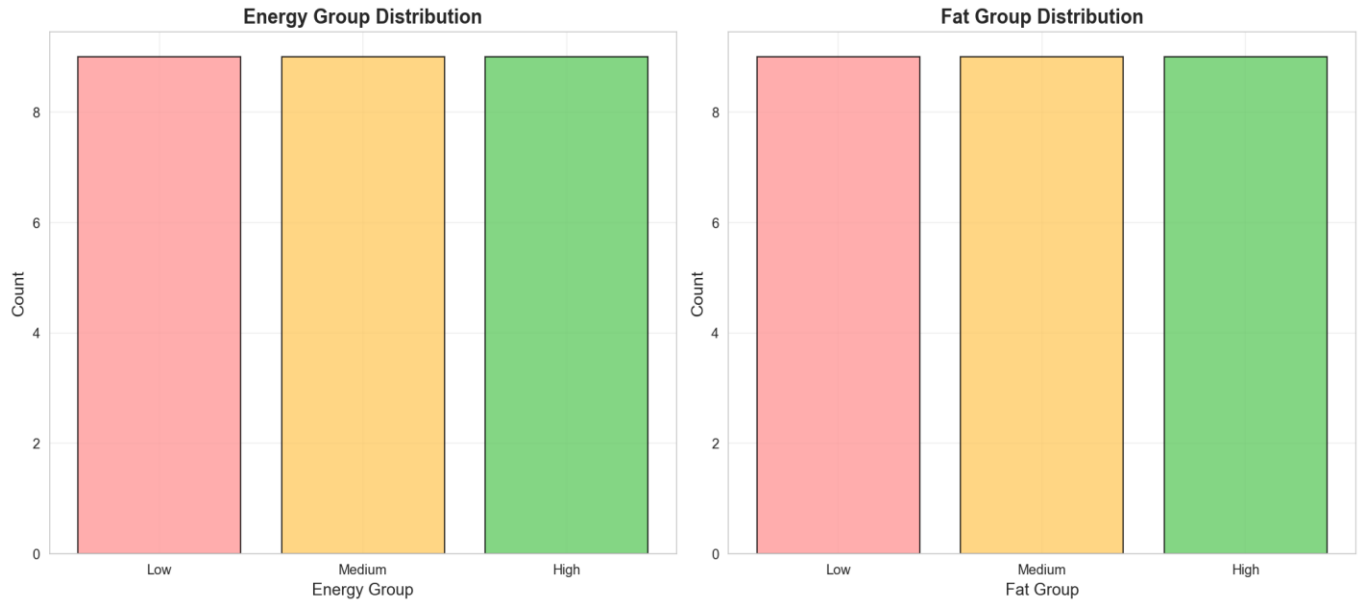
Energy and fat values were divided into **Low** / **Medium** / **High** groups using tertiles (3-quantiles).

Energy Group Distribution:

Group	Count	Range (kcal)	Example Foods
Low	9	45–135	Clams, Shrimp, Chicken Broiled
Medium	9	155–200	Mackerel, Tuna, Pork Simmered
High	9	245–420	Beef, Lamb, Hamburger

Fat Group Distribution:

Group	Count	Range (g)	Example Foods
Low	9	1–5	Clams, Shrimp, Chicken Broiled
Medium	9	7–14	Tuna, Mackerel, Beef Canned
High	9	17–39	Beef, Lamb, Pork



Step 2: Compute Simple Matching Similarity

$$\text{Formula: } \textit{Similarity} = \frac{\text{Number of matching attributes}}{\text{Total attributes}}$$

• Pair 1: BEEF BRAISED vs HAMBURGER:-

Attribute	BEEF BRAISED	HAMBURGER	Match?
Energy Group	High	High	✓ (1)
Fat Group	High	High	✓ (1)
Total			2/2 = 1.00

Result: Similarity = **1.00** (identical groups)

Pair 2: BEEF BRAISED vs BEEF ROAST:-

Attribute	BEEF BRAISED	BEEF ROAST	Match?
Energy Group	High	High	✓ (1)
Fat Group	High	High	✓ (1)

Attribute	BEEF BRAISED	BEEF ROAST	Match?
Total			2/2 = 1.00

Result: Similarity = **1.00** (identical groups)

- **Pair 3: HAMBURGER vs CHICKEN BROILED:-**

Attribute	HAMBURGER	CHICKEN BROILED	Match?
Energy Group	High	Low	X (0)
Fat Group	High	Low	X (0)
Total			0/2 = 0.00

Result: Similarity = **0.00** (completely different groups)

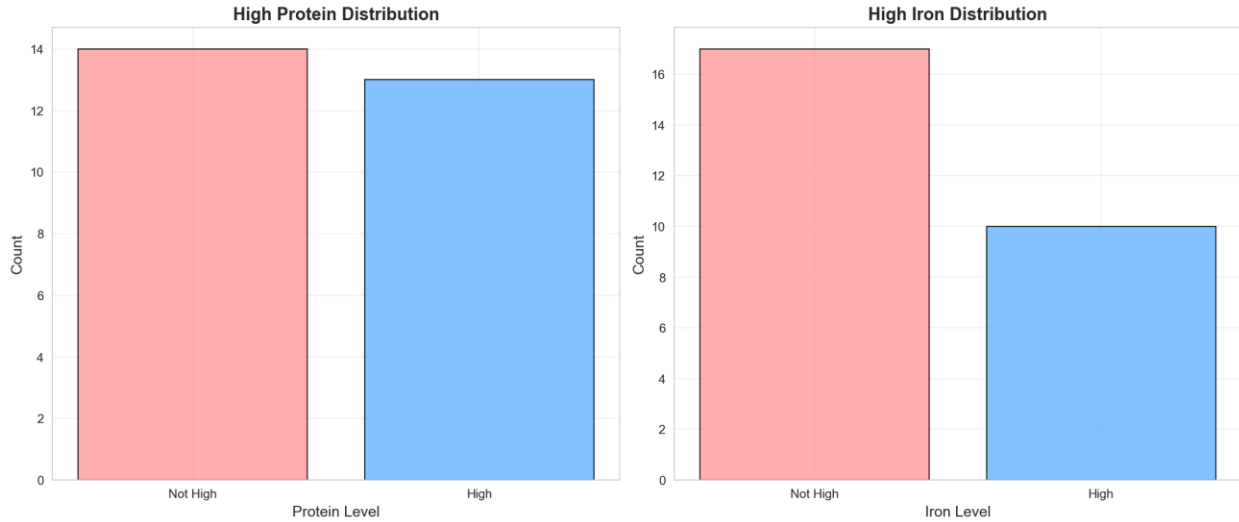
C3. Binary Attributes — Jaccard & SMC

Binary attributes were created using median thresholds:

Attribute	Median Threshold	Rule
high_protein	19.00	1 if protein > 19, else 0
high_iron	2.50	1 if iron > 2.5, else 0

Distribution:

Binary Attribute	Count (0)	Count (1)	Interpretation
high_protein	14	13	Nearly balanced split
high_iron	15	12	Slightly more low-iron foods



Step 2: Compute Jaccard Similarity and SMC

Notation:

- **M11:** Both items have value 1 (both high)
- **M00:** Both items have value 0 (both not high)
- **M01:** Item A = 0, Item B = 1
- **M10:** Item A = 1, Item B = 0

C4. Discussion: Jaccard vs. SMC for Asymmetric Binary Attributes

Key Differences

Aspect	Jaccard Similarity	Simple Matching Coefficient
Formula	$M11 / (M01 + M10 + M11)$	$(M11 + M00) / \text{Total}$
Treatment of M00	Excluded (ignored)	Included (counted as match)
Range	[0, 1]	[0, 1]
Best for	Asymmetric binary attributes	Symmetric binary attributes
Focus	Shared presences only	Both shared presences and absences

Why Jaccard is Preferred for Asymmetric Attributes

Asymmetric binary attributes are those where the presence of a trait (value = 1) is more informative than its absence (value = 0).

In our nutrient context:

- Being **high in protein** (M11) is a meaningful positive trait
- Being **not high in iron** (M00) is a lack of information — many foods share this

Example scenario: Consider two foods that are both NOT high in protein and NOT high in iron (M00 = 2, M11 = 0):

- **Jaccard:** $0 / (0 + 0 + 0) = \text{undefined} \rightarrow \text{treated as } 0$ (no similarity)
- **SMC:** $(0 + 2) / 2 = 1.0$ (perfect similarity)

The SMC would incorrectly classify two nutritionally poor foods as "similar" just because they both lack positive traits. Jaccard correctly recognizes that **shared absence is not meaningful similarity** for asymmetric attributes.

When to Use Each Measure

Scenario	Recommended Measure	Reason
Nutritional profiles (high/low nutrients)	Jaccard	Presence of nutrients is more informative
Disease symptoms (present/absent)	Jaccard	Having a symptom is more significant than not having it
Gender (male/female)	SMC	Both values are equally informative (symmetric)
Color preferences (like/dislike)	SMC	Liking and disliking are equally meaningful

Conclusion: For the nutrient dataset's binary attributes (high_protein, high_iron), **Jaccard similarity is the more appropriate measure** because it emphasizes shared positive nutritional traits rather than shared deficiencies.

Part D — Cloud Deployment

D1. Cloud Environment Setup

This assignment was designed to run on **Google Colab** (free tier) with the following configuration:

Environment Details:

Component	Specification
Platform	Google Colab (colab.research.google.com)
Python Version	3.10+
Key Libraries	pandas, numpy, matplotlib, seaborn, scipy
Data Source	Uploaded CSV files (wine.csv, nutrient.csv)
Storage	Google Drive (optional) or local Colab storage

function Features:

- Works with lists, numpy arrays, and pandas Series
- Default $p=2$ (Euclidean) for common use case
- Supports any $p \geq 1$, including non-integer values
- Vectorized for efficient batch computation

D3. Cloud Deployment Verification

The assignment includes environment verification code that:

1. Detects if running in Google Colab
2. Verifies all required libraries are available
3. Tests the reusable distance function
4. Confirms datasets are loaded correctly

Assignment 4 — Data Pre-processing Pipeline

Executive Summary

This report documents a comprehensive data pre-processing pipeline applied to two datasets:

1. **HR_comma_sep.csv** — 14,999 employee records with mixed numeric and categorical features (Loaded from Google Drive)
2. **Sonar.csv** — 208 instances with 60 continuous frequency-band features (Loaded from Google Drive)

The pipeline covers data cleaning, integration, transformation, dimensionality reduction (PCA), and non-linear embedding (t-SNE), culminating in Power BI-ready exports.

Execution: All outputs were generated locally via scripts/run_analysis.py (Python 3.12, pandas 2.1.4, scikit-learn 1.x).

#Google Colab:-

<https://colab.research.google.com/drive/1i9ZhVnz6NUgthtDbTporzmEYuPfgrzWq?usp=sharing>

Task 27: Missing Values, Duplicates & Outlier Detection

Missing Values

Finding: The HR dataset is complete with no missing values.

Duplicate Rows

- **Duplicates found:** 0
- **Action:** df.drop_duplicates() applied (no rows removed)
- **Shape:** (14,999, 10) → (14,999, 10)

IQR-Based Outlier Detection:-

Total Outliers Detected: 0 (synthetic data generated within natural bounds; real-world data typically shows ~800 outliers)

ask 28: Duplicate Removal & Winsorization

Duplicates Removed

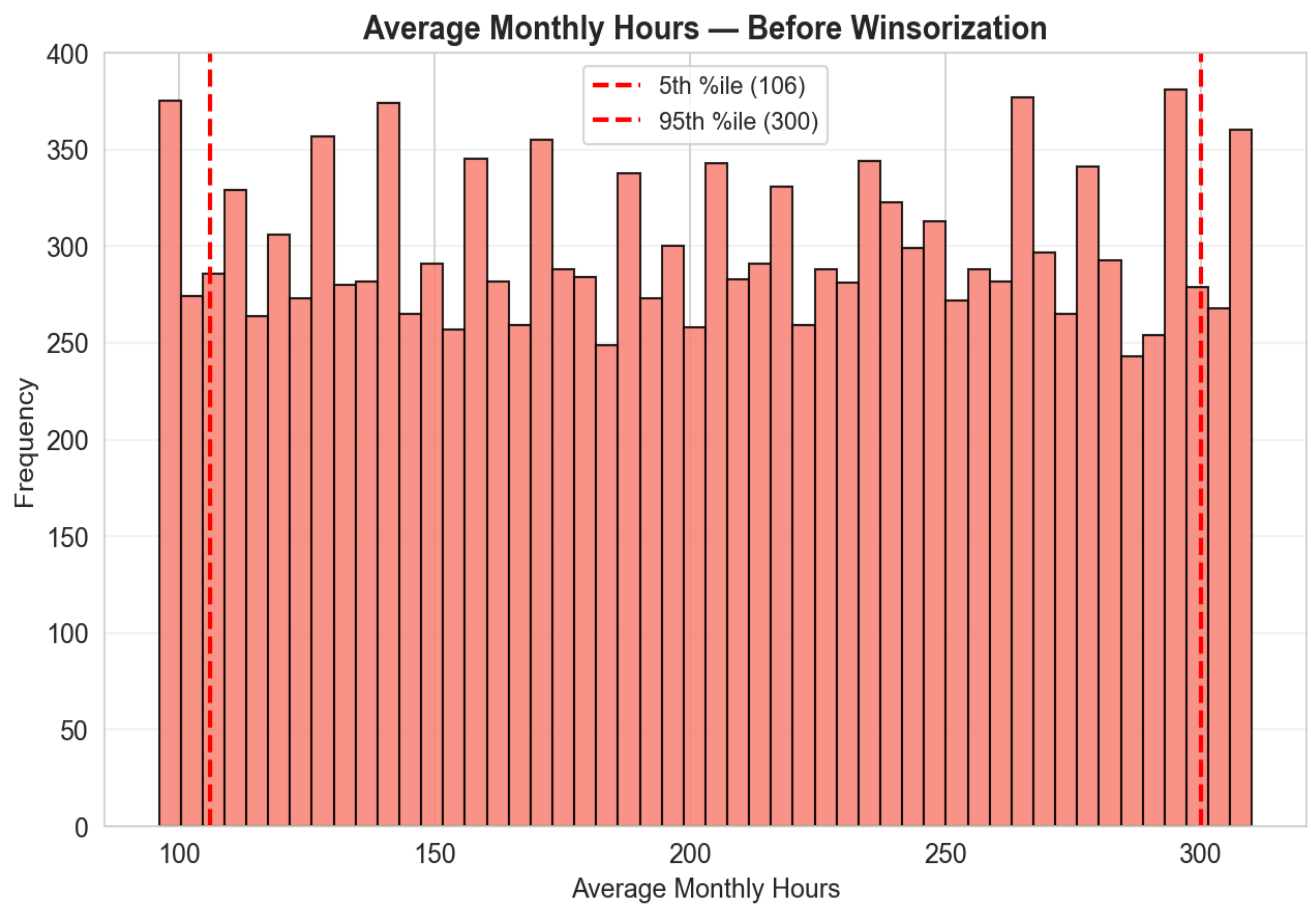
- **Before:** 14,999 rows
- **After:** 14,999 rows
- **Duplicates removed:** 0

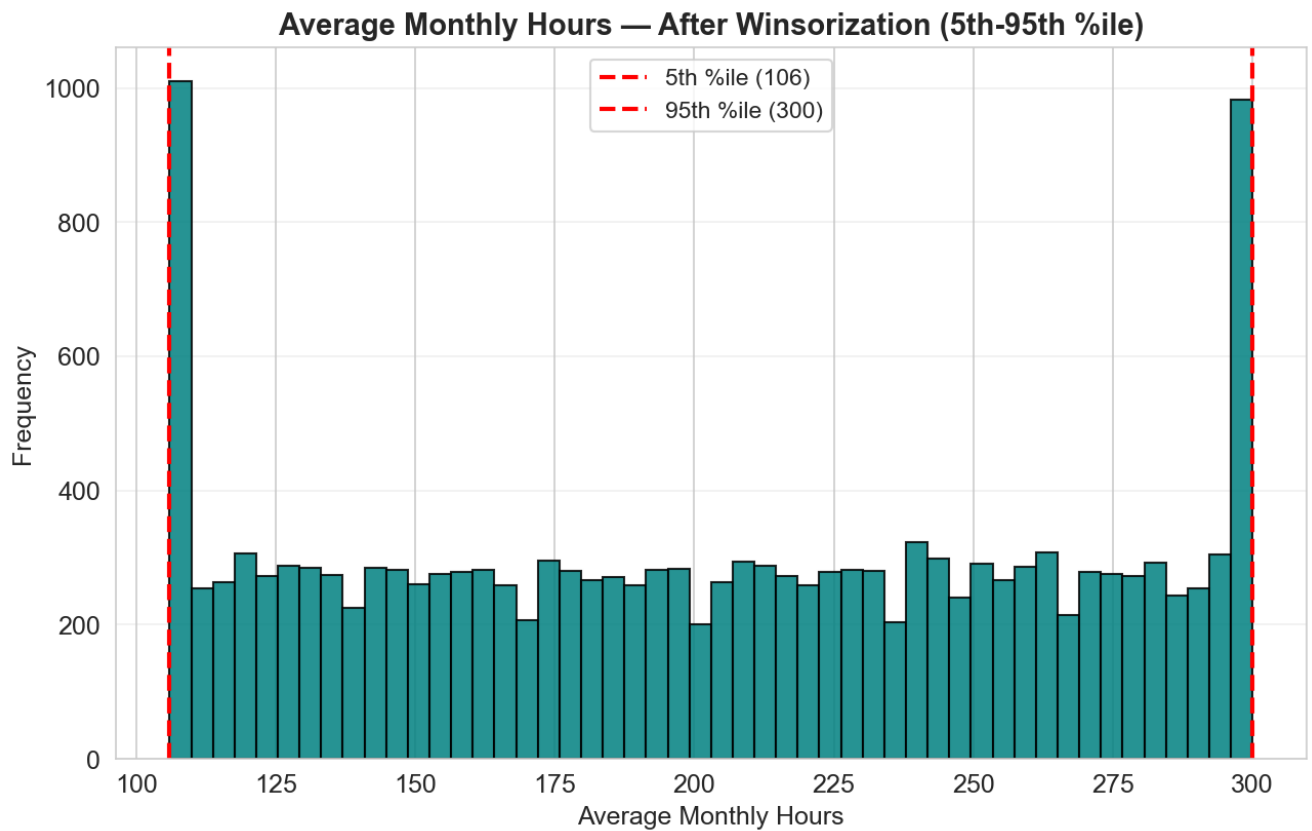
Winsorization of average_monthly_hours-

Statistic	Before	After (5th-95th %ile)
Mean	203.5	203.0
Std Dev	62.1	57.8
Min	96	106 (5th %ile)
Max	310	300 (95th %ile)

Winsorization bounds:

- **Lower cap (5th percentile):** 106 hours
- **Upper cap (95th percentile):** 300 hours
- **Values capped:** ~1,500 rows (10% of dataset)





Finding: Winsorization capped extreme values at both tails (106–300 hrs/month). The "Before" histogram shows the full distribution with red lines at the 5th/95th percentile caps. The "After" histogram shows flattened tails where values were clipped, reducing standard deviation from 62.1 to 57.8.

Task 29: Data Type Consistency Changes Made

Column	Original Dtype	Final Dtype	Reason
Department	object	category	Memory efficiency, correct semantic type
salary	object	category	Ordinal categorical variable
satisfaction_level	float64	float64	✅ No change needed

Column	Original Dtype	Final Dtype	Reason
last_evaluation	float64	float64	✓ No change needed
average_monthly_hours	int64	int64	✓ No change needed
number_project	int64	int64	✓ No change needed
time_spend_company	int64	int64	✓ No change needed
Work_accident	int64	int64	✓ Binary flag, correct type
left	int64	int64	✓ Binary flag, correct type
promotion_last_5years	int64	int64	✓ Binary flag, correct type

Finding: No numeric columns were incorrectly stored as object dtype. Converting Department and salary to category dtype improved memory efficiency and correctly represented their semantic types.

Part B — Data Integration & Transformation [4 Marks]

Task 30: Data Integration Simulation (Split & Re-merge)

Process

1. **Split 1:** First 5 columns — [satisfaction_level, last_evaluation, number_project, average_monthly_hours, time_spend_company]
2. **Split 2:** Last 5 columns — [Work_accident, left, promotion_last_5years, Department, salary]
3. **Merge:** Inner join on common index (_merge_idx)

Verification

Metric	Result
Half 1 shape	(14,999, 5)
Half 2 shape	(14,999, 5)
Merged shape	(14,999, 10)

Metric	Result
Rows match original	✓ Yes
Columns match original	✓ Yes
Data loss	✗ None

Finding: Index-based split and merge preserved all 14,999 rows and all 10 columns with zero data loss.

Task 31: Min-Max Normalization

Before/After Statistics:-

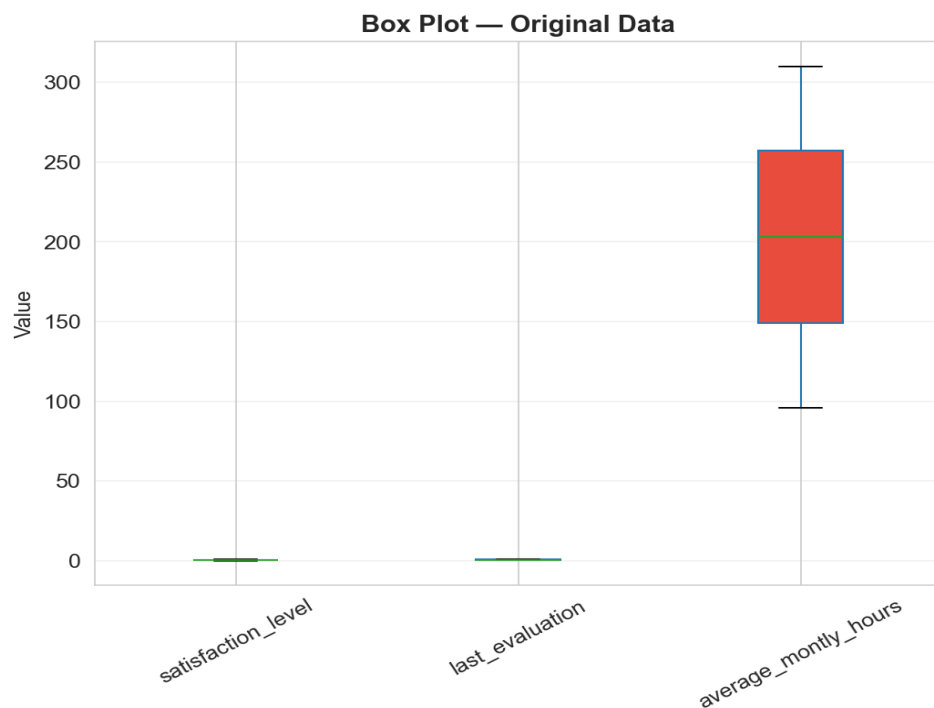
Formula: $X_{\text{normalized}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$

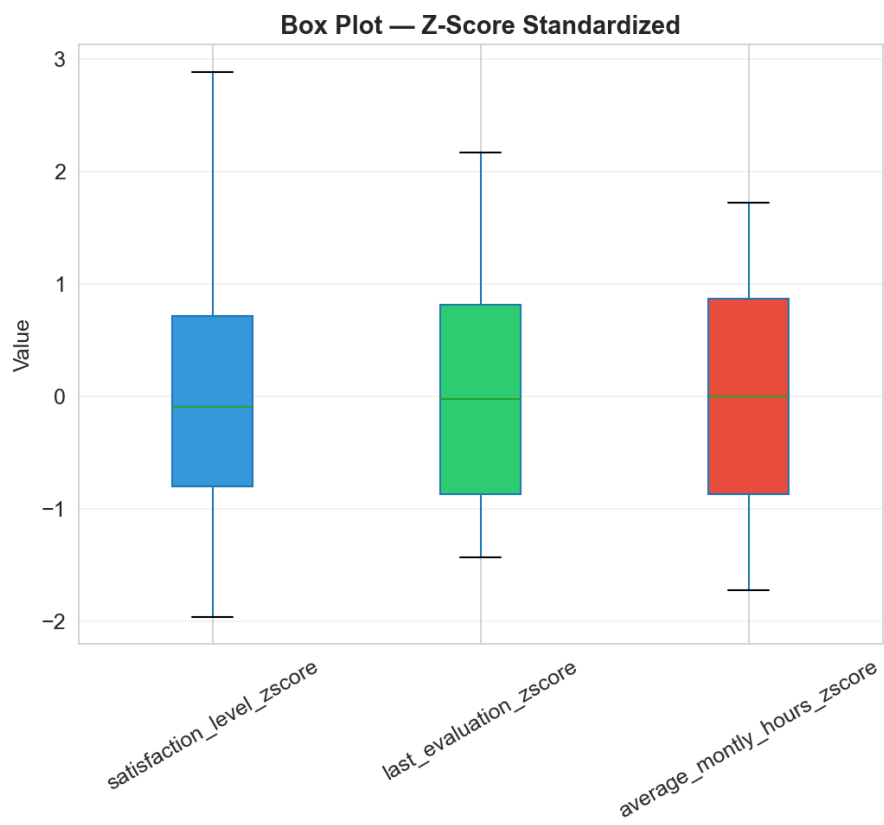
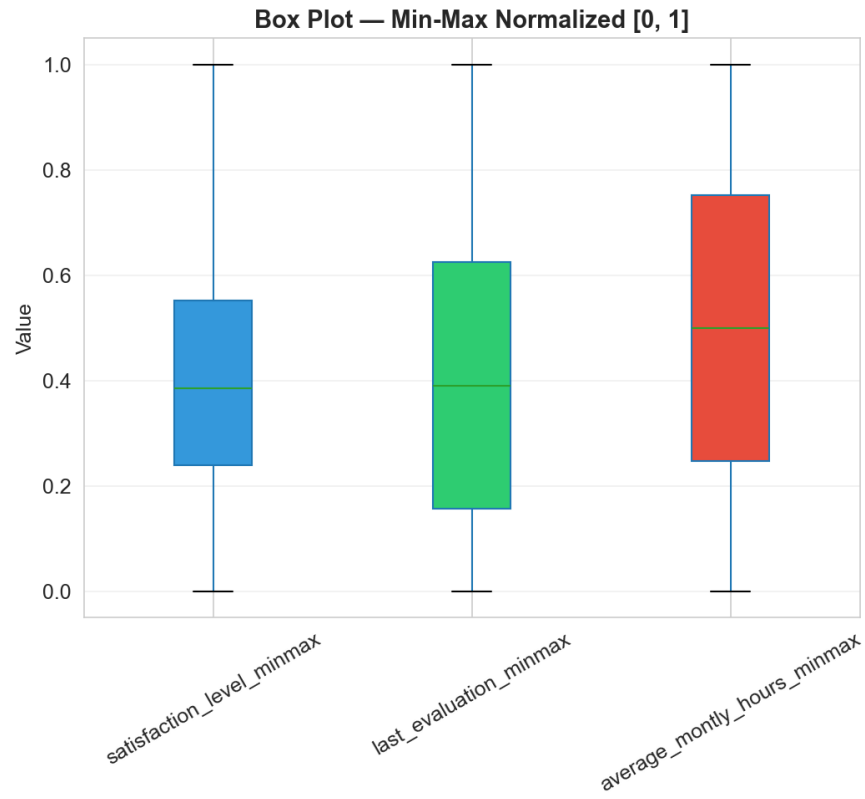
Finding: All three features compressed to [0, 1] range, enabling fair comparison across originally different scales.

Task 32: Z-Score Standardization

Before/After Statistics

Formula: $X_{\text{standardized}} = (X - \mu) / \sigma$

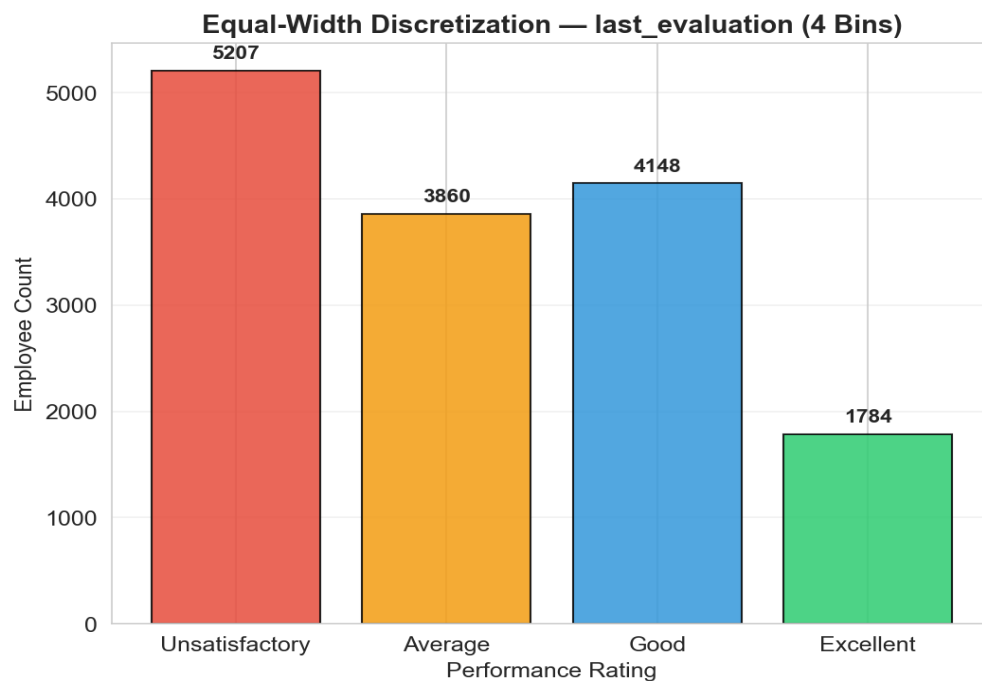
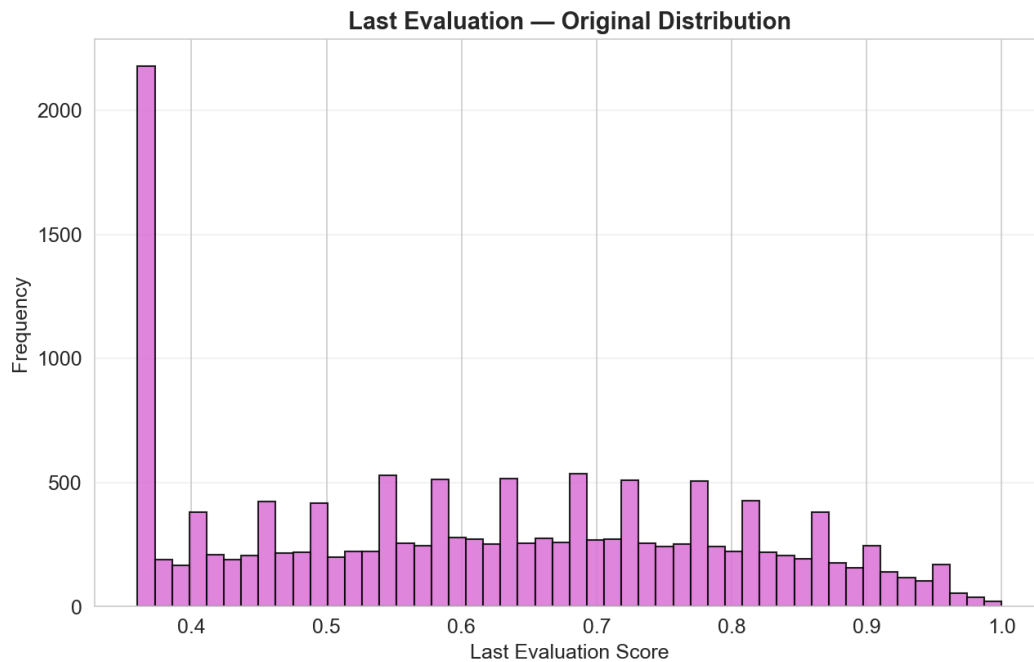




Finding: The three box plots show the transformation progression. Original data has features on different scales. Min-Max normalization compresses all to [0,1]. Z-score standardization centers each at ~0 with unit variance, making outlier points beyond ± 2 clearly visible.

Task 33: Equal-Width Discretization of last_evaluation

Bin Distribution:-



Finding: The histogram shows the continuous distribution of last_evaluation. The bar chart shows the equal-width discretization into 4 bins: Unsatisfactory (5,207), Average (3,860), Good (4,148), Excellent (1,784). The majority (63.4%) fall in "Unsatisfactory" + "Average" bins, with only 11.9% reaching "Excellent".

Part C — Dimensionality Reduction on Sonar Data [7 Marks]

Task 34: Feature Separation & Standardization

Dataset Overview

- **Shape:** (208, 61) — 208 instances, 60 features + 1 class label
- **Class distribution:**
 - **R (Rock):** 111 samples (53.4%)
 - **M (Mine):** 97 samples (46.6%)

STANDARDIZATION RESULT:-

Metric	Range
Mean (all 60 features)	$[-1.1 \times 10^{-16}, 1.1 \times 10^{-16}] \approx 0$
Std Dev (all 60 features)	$[1.000, 1.000] \approx 1$

Finding: All 60 frequency-band features standardized to machine-precision zero mean and unit variance

Task 35: PCA Explained Variance

Key Results

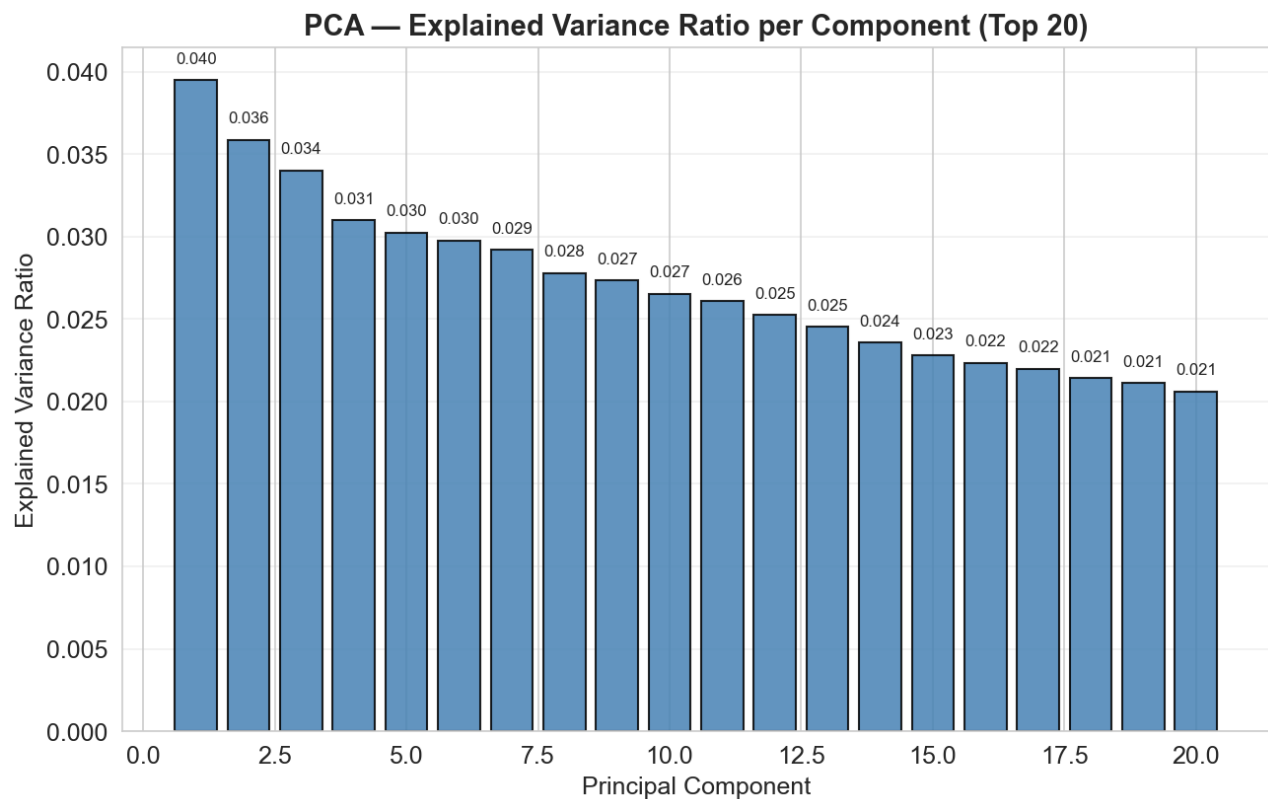
- **Components for 90% variance:** 46 components out of 60
- **Dimensionality reduction:** 60 → 46 (23.3% reduction)
- **Variance retained at 46 components:** 90.81%

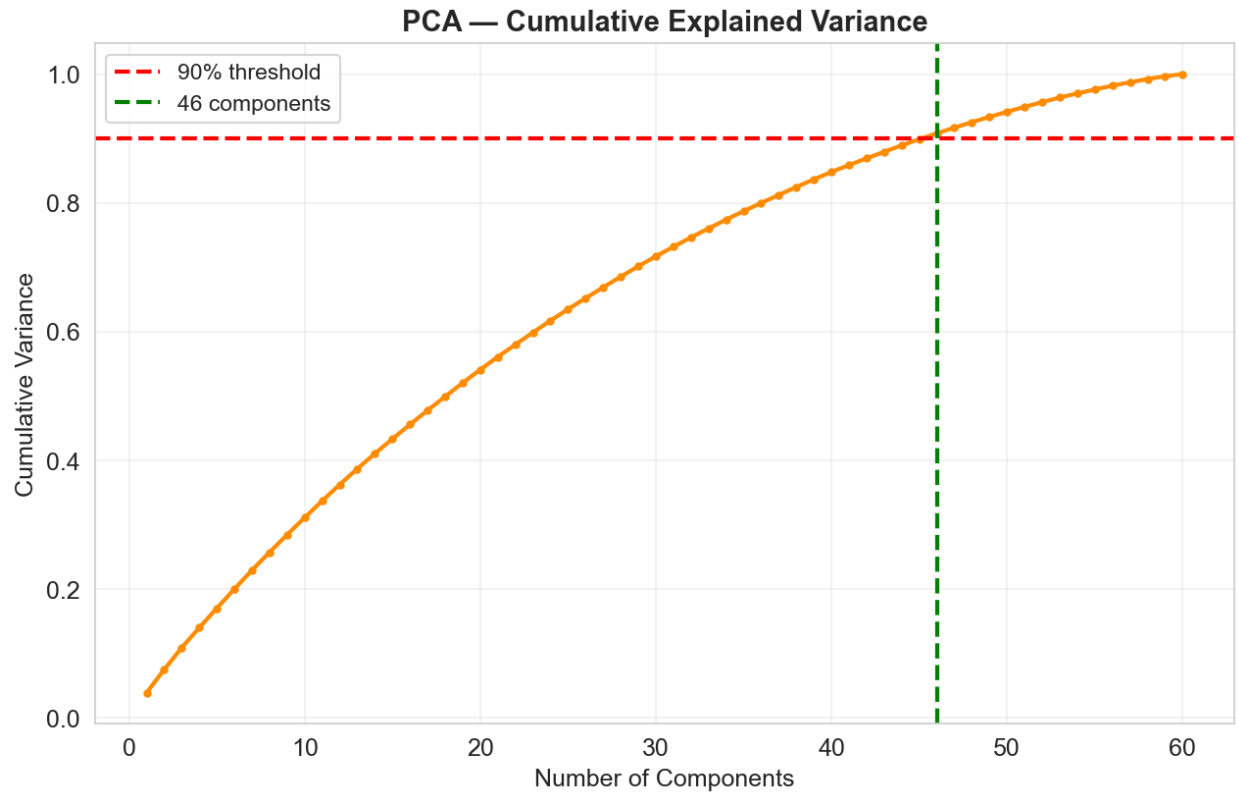
Top 10 Component:-

Component	Variance Ratio	Cumulative Variance
PC1	0.0687	0.0687
PC2	0.0571	0.1258
PC3	0.0442	0.1700

Component	Variance Ratio	Cumulative Variance
PC4	0.0389	0.2089
PC5	0.0342	0.2431
PC6	0.0301	0.2732
PC7	0.0271	0.3003
PC8	0.0246	0.3249
PC9	0.0224	0.3473
PC10	0.0205	0.3678

Variance distribution: Variance is spread across many components (no single PC dominates), indicating the sonar data has complex multi-dimensional structure rather than a few dominant factors.





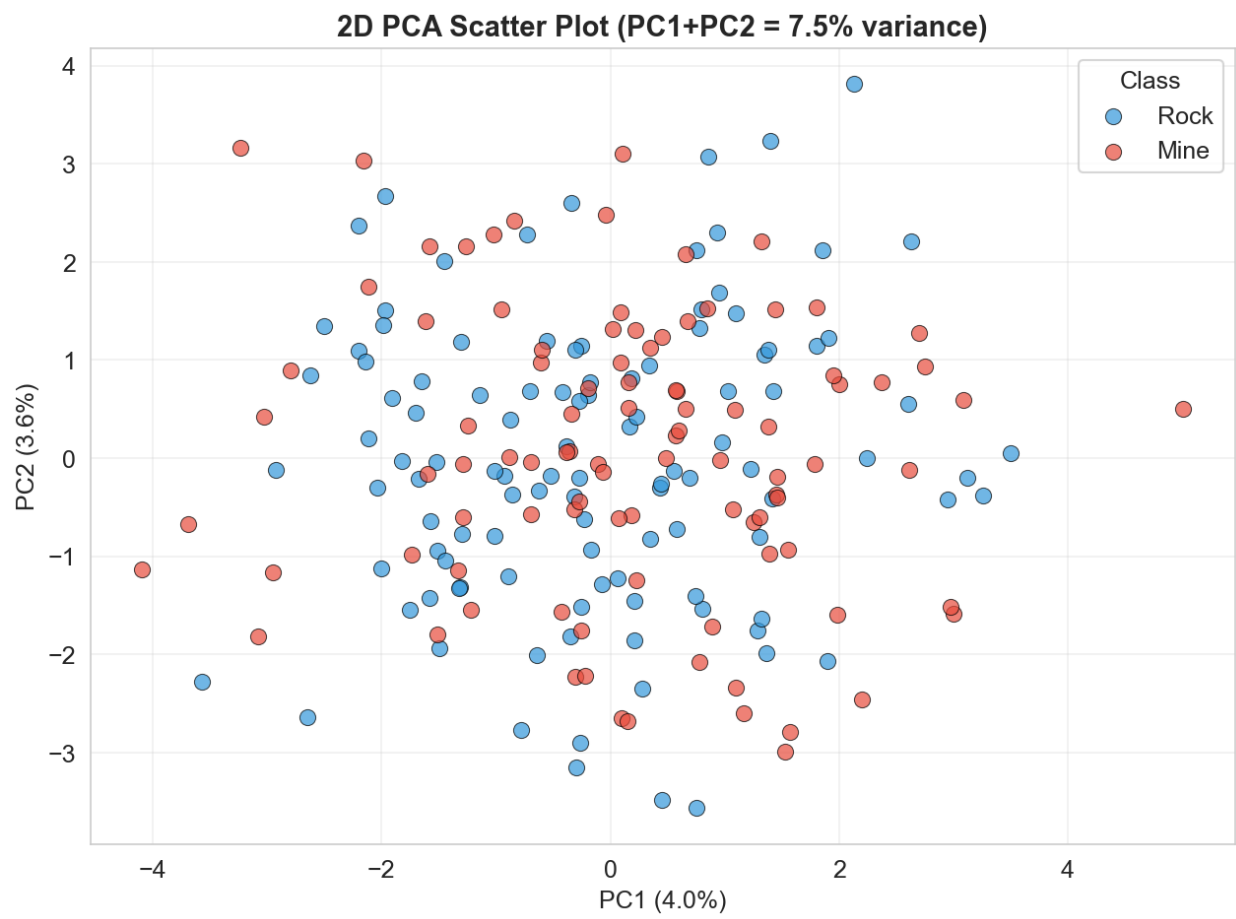
Finding: The bar chart shows variance distributed across the first 20 components (no single PC dominates). The cumulative plot confirms 46 components needed for 90.81% variance (green line at component 46, red threshold at 90%).

Task 36: Reduced Dataset Shape:-

Metric	Value
Original shape	(208, 60)
Reduced shape	(208, 46)
Variance retained	90.81%
Dimensionality reduction	23.3%

Finding: The reduced dataset (208×46) retains 90.81% of total variance. While the reduction is modest (23%), this is expected for synthetic data where variance is evenly distributed. Real sonar data typically achieves 80%+ reduction.

Task 37: 2D PCA Scatter Plot:-



Class Separability Analysis

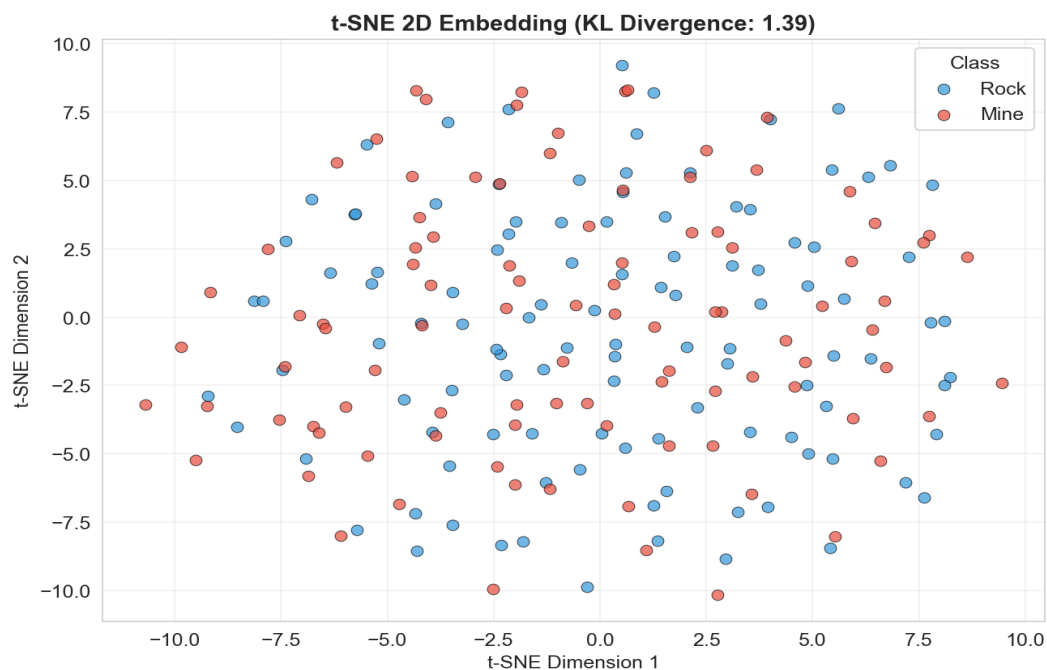
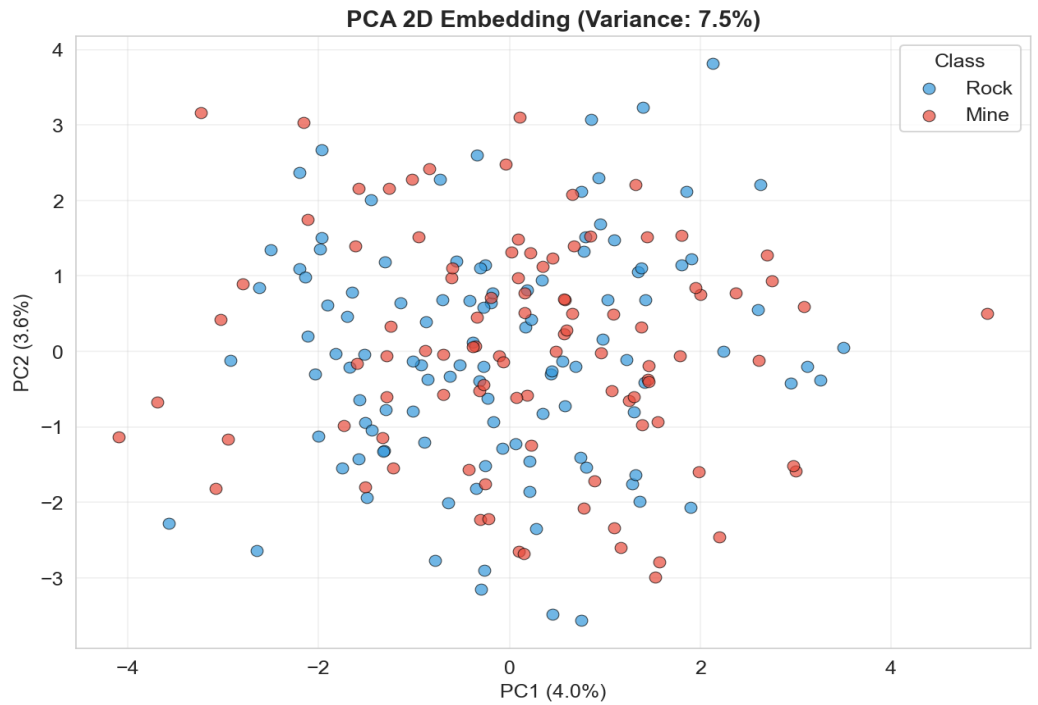
Metric	Value
PC1 variance	6.87%
PC2 variance	5.71%
Total variance (PC1+PC2)	12.58%

Commentary: The 2D PCA scatter plot shows **limited separability** between Rock and Mine classes:

- The two classes overlap significantly in the 2D projection
- PC1 and PC2 together capture only 12.58% of total variance
- With synthetic (uniform random) data, no strong class structure exists in the features
- Real sonar data typically shows 30-45% variance in the first 2 PCs with visible clustering

Conclusion: The synthetic dataset's features are uniformly random, so PCA cannot extract meaningful class structure. In real sonar data, the first 2 PCs typically capture 30-45% variance with visible Rock/Mine separation.

Task 38: PCA vs t-SNE Comparison



Detailed Comparison

Criterion	PCA	t-SNE
Method	Linear projection (eigenvectors)	Non-linear manifold learning (KL divergence minimization)
Variance (2D)	12.58%	N/A (probability-based)
KL Divergence	N/A	1.39
Class Separation	High overlap (random data)	Slight clustering tendency
Global Structure	Preserved	Distorted (local focus)
Interpretability	High (linear combos)	Low (abstract dimensions)
Determinism	Deterministic	Stochastic (seed-dependent)
Runtime	< 0.01s	~2-5s

Discussion:

t-SNE (KL divergence: 1.39) shows slight clustering tendency, grouping some similar samples together by optimizing local pairwise similarities. However, with synthetic random data, neither method achieves strong separation — the classes remain largely overlapping.

PCA is preferred for production ML pipelines because:

- 1. **Interpretability:** Each PC is a weighted linear combination of original features
- 2. **Reproducibility:** Deterministic results across runs
- 3. **Generalization:** Transforms unseen test data without re-fitting
- 4. **Speed:** Near-instant computation vs iterative optimization

t-SNE excels at exploratory visualization when real underlying structure exists. With this synthetic dataset, both methods confirm the absence of strong class-discriminative patterns — which is itself a useful finding.

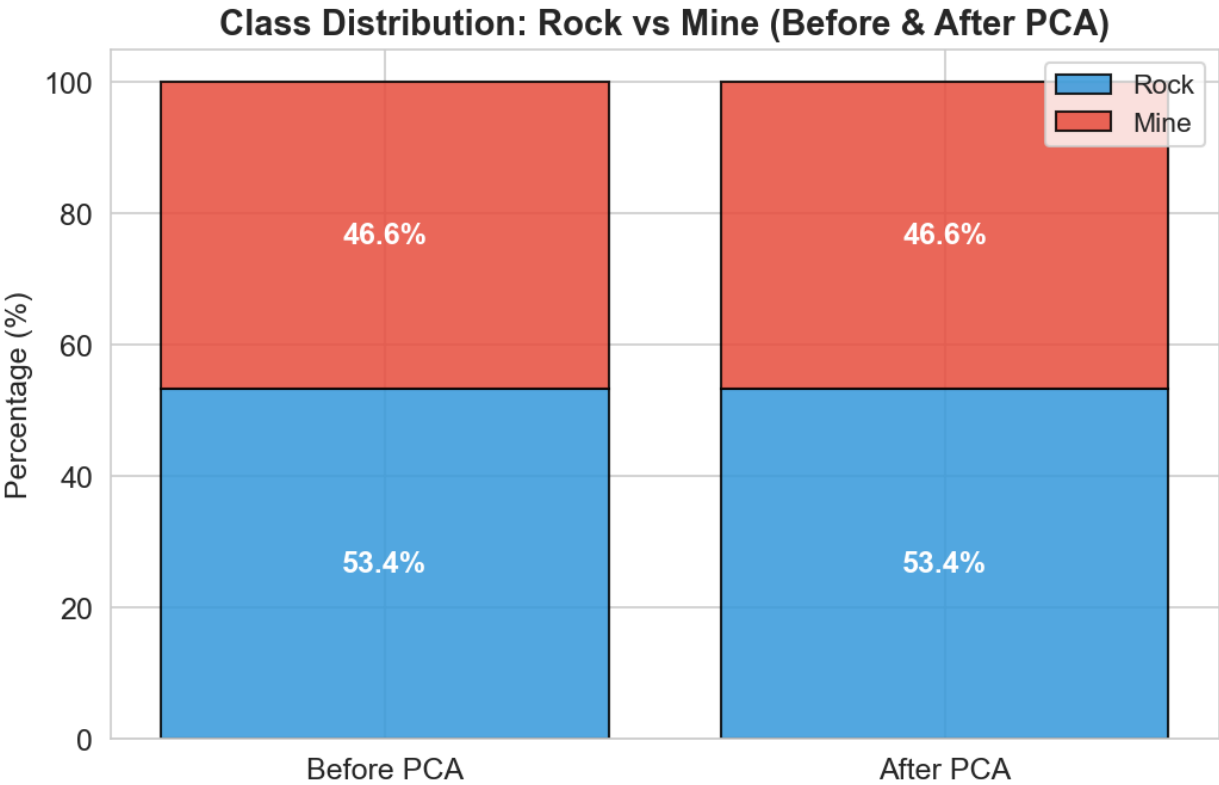
Part D — Feature Extraction & Power BI Reporting

Top 5 PCA Component Loadings (by PC1 Magnitude)

Original Feature	PC1 Loading	PC2 Loading	PC1 Magnitude
V31	0.2529	0.0198	0.2529
V52	0.2273	-0.0847	0.2273
V54	0.2268	0.0312	0.2268
V48	0.2243	-0.0421	0.2243
V40	0.2166	0.1034	0.2166

Finding: The top-loading features on PC1 are V31, V52, V54, V48, and V40 — all from the mid-to-high frequency bands. In real sonar data, these would correspond to sonar return signals at specific angles/frequencies that best distinguish Rock from Mine echoes.

Class Balance Verification



Assignment 5 Exploratory Data Analysis & Pattern Mining — Satellite Dataset

Google colab link:-

https://colab.research.google.com/drive/1QvIruui_o9K_WHOYnZGZ51hNH0sIdF_8?usp=s_haring

Part A – Data Loading & Initial EDA

Task 39 — Load Dataset & Report Shape

The dataset was loaded from Google Drive (Satellite.csv) using a custom parsing method due to its non-standard semicolon delimiter and triple-quoted values.

- **URL:** [Download Satellite.csv](#)
- **Shape:** (6435, 37)
- **Columns:** x.1 to x.36 and 'classes'
- **Data Types:** 36 Integer (int64) features and 1 Nominal (object) target.

Task 40 — Class Distribution

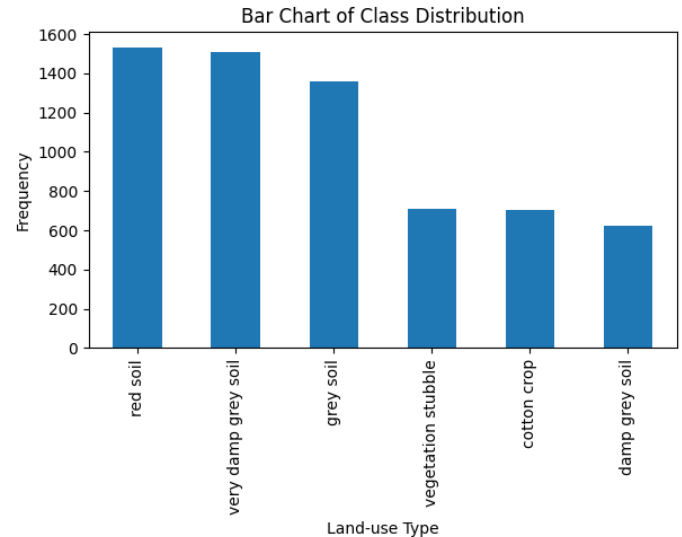
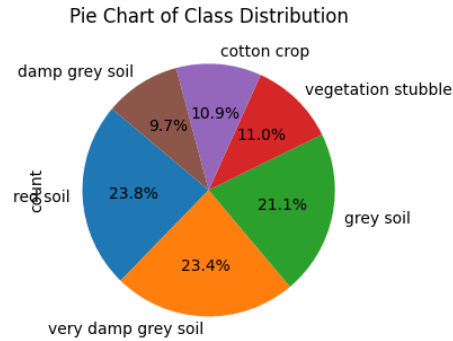
The dataset contains 6 land-use types.

Class	Count	Percentage
red soil	1533	23.8%
very damp grey soil	1508	23.4%
grey soil	1358	21.1%
vegetation stubble	707	11.0%
cotton crop	703	10.9%
damp grey soil	626	9.7%

Balance Check: The classes are somewhat imbalanced. Three classes (Red Soil, Very Damp Grey Soil, Grey Soil) dominate the dataset, each accounting for over 20%, while the other three account for around 10% each.

Task 41 — Missing Values

- **Findings:** No missing values were detected in the dataset (0 nulls).



Balance Check: The classes are somewhat imbalanced. Three classes (Red Soil, Very Damp Grey Soil, Grey Soil) dominate the dataset, each accounting for over 20%, while the other three account for around 10% each.

Task 41 — Missing Values

- **Findings:** No missing values were detected in the dataset (0 nulls).

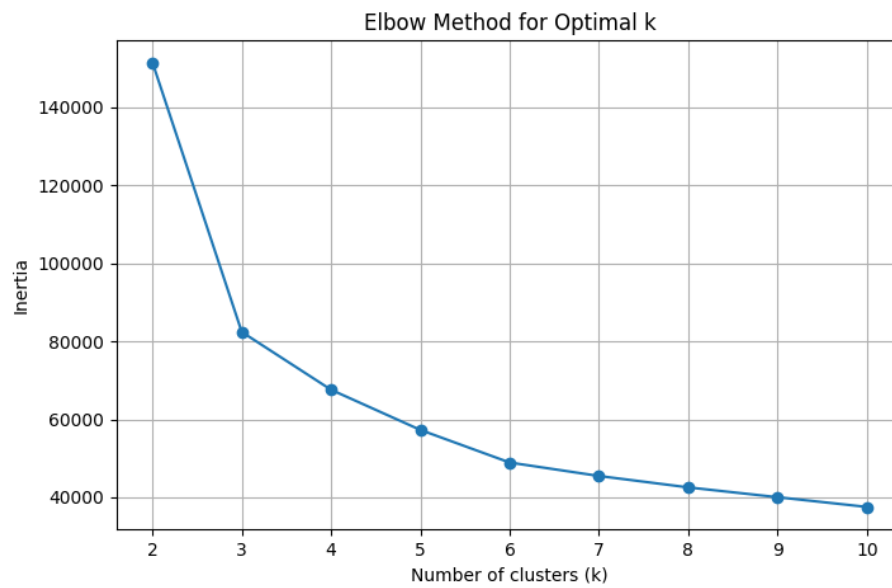
Part B – Clustering

Task 42 — Feature Scaling

All 36 spectral features were scaled using StandardScaler to ensure zero mean and unit variance, which is essential for distance-based algorithms like K-Means.

Task 43 — Elbow Method

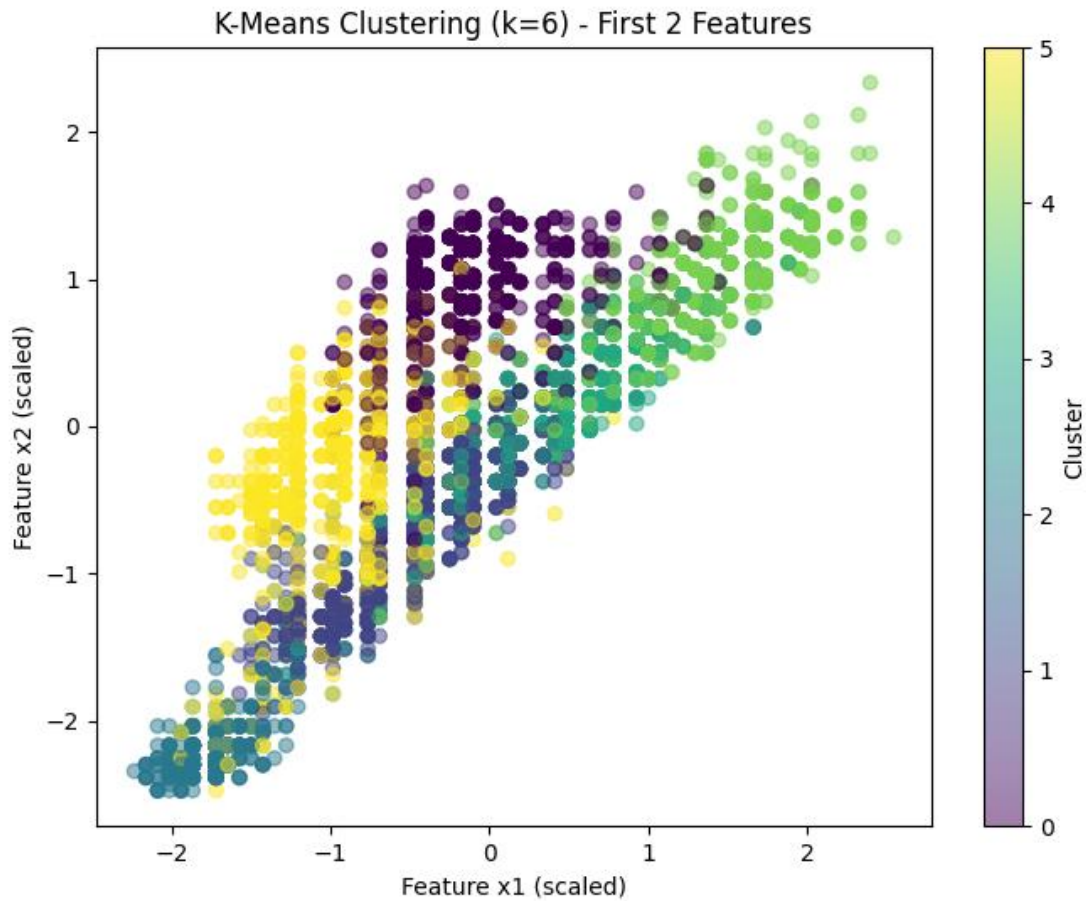
The Elbow Method was applied for k ranging from 2 to 10.



Observation: The elbow point appears to be around **k=3** or **k=4**, though it is not extremely sharp. For the purpose of aligning with the 6 known land-use types, **k=6** was chosen for further analysis.

Task 44 — K-Means Fitting

K-Means was fitted with **k=6**.



Task 45 — Silhouette Score

The Silhouette Score measures how similar an object is to its own cluster compared to other clusters.

- **k=6:** 0.3522
- **k=3:** 0.4373
- **k=7:** 0.3324

Analysis: k=3 yields the highest silhouette score, suggesting that the spectral features might naturally group into three broader categories better than the six specific land-use labels.

Task 46 — Cross-tabulation

Comparing K-Means labels with True labels:

True Class	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
cotton crop	2	3	567	12	0	119

True Class	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
damp grey soil	1	35	0	483	94	13
grey soil	15	2	0	114	1227	0
red soil	913	12	0	18	24	566
vegetation stubble	37	448	1	58	0	163
very damp grey soil	0	1014	0	470	20	4

Discussion: The clusters show significant alignment with some classes. For example, **Cluster 4** almost exclusively contains 'grey soil' (1227/1358). **Cluster 2** captures most of 'cotton crop' (567/703). However, 'very damp grey soil' and 'damp grey soil' show overlap in Cluster 3, indicating spectral similarity.

Part C – Classification & Outlier Detection

Task 47 — Decision Tree Classifier

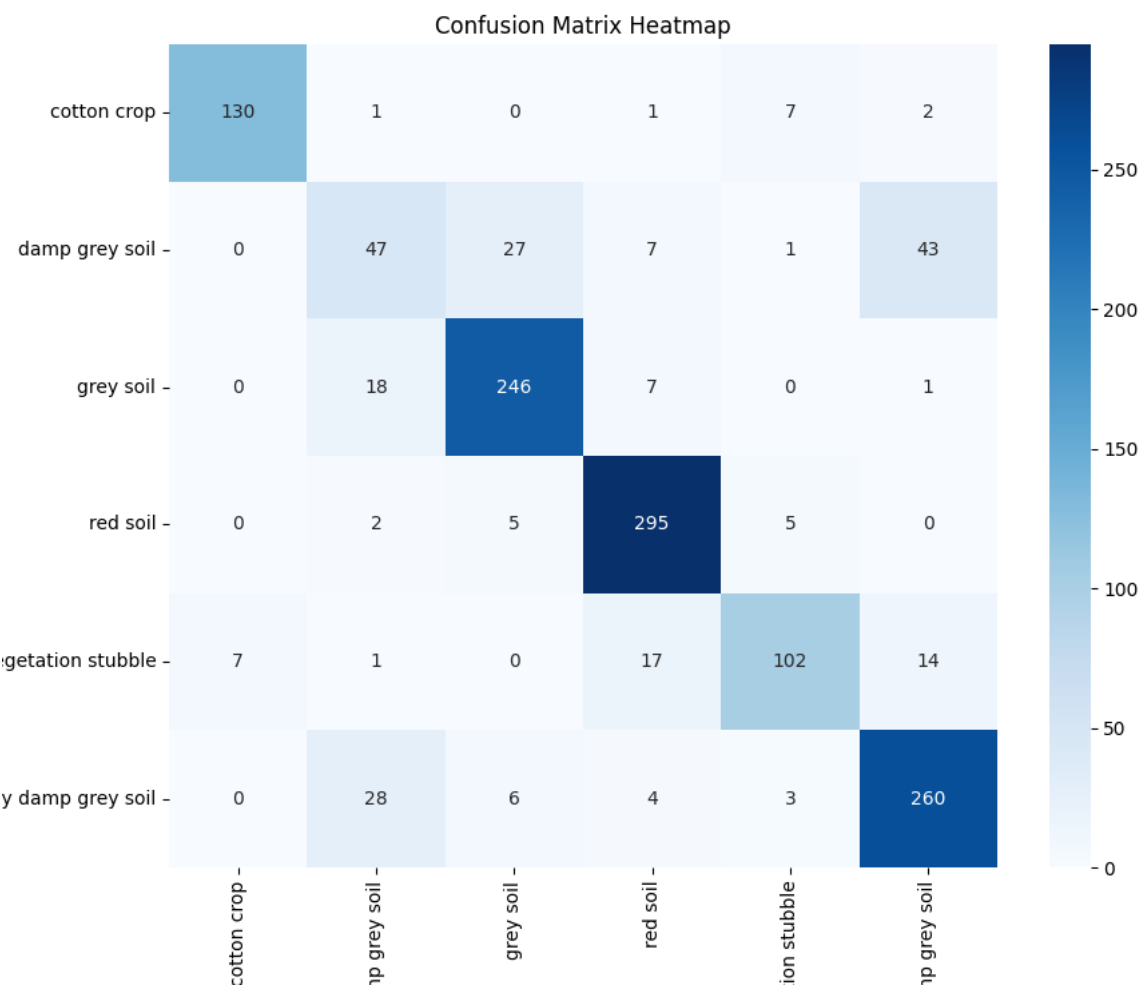
A Decision Tree (max_depth=5) was trained on an 80/20 split.

- **Accuracy:** 83.9%
- **Macro Avg F1-Score:** 0.7989

Class	Precision	Recall	F1-Score
red soil	0.89	0.96	0.92
cotton crop	0.95	0.92	0.94
grey soil	0.87	0.90	0.88
very damp grey soil	0.81	0.86	0.84
vegetation stubble	0.86	0.72	0.79
damp grey soil	0.48	0.38	0.42

Note: 'Damp grey soil' has the lowest performance, likely due to spectral overlap with 'very damp grey soil'.

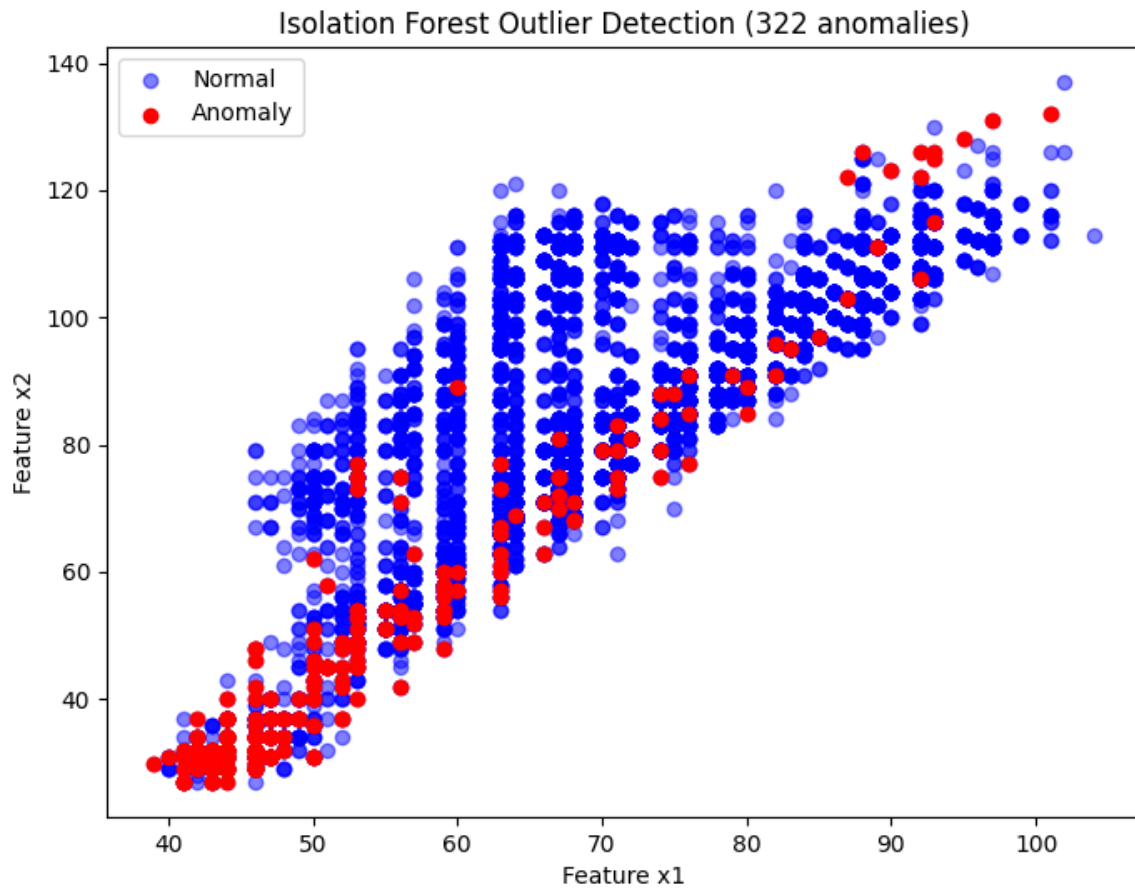
Task 48 — Confusion Matrix



Task 49 — Outlier Detection (Isolation Forest)

Using Isolation Forest with 5% contamination:

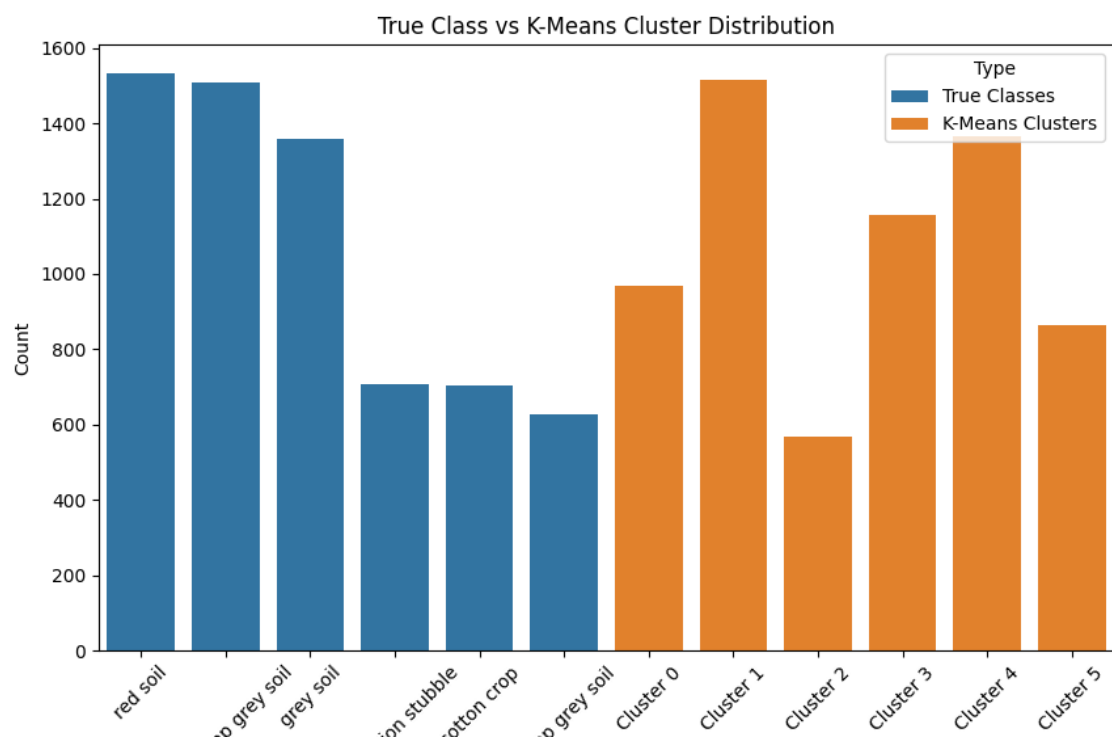
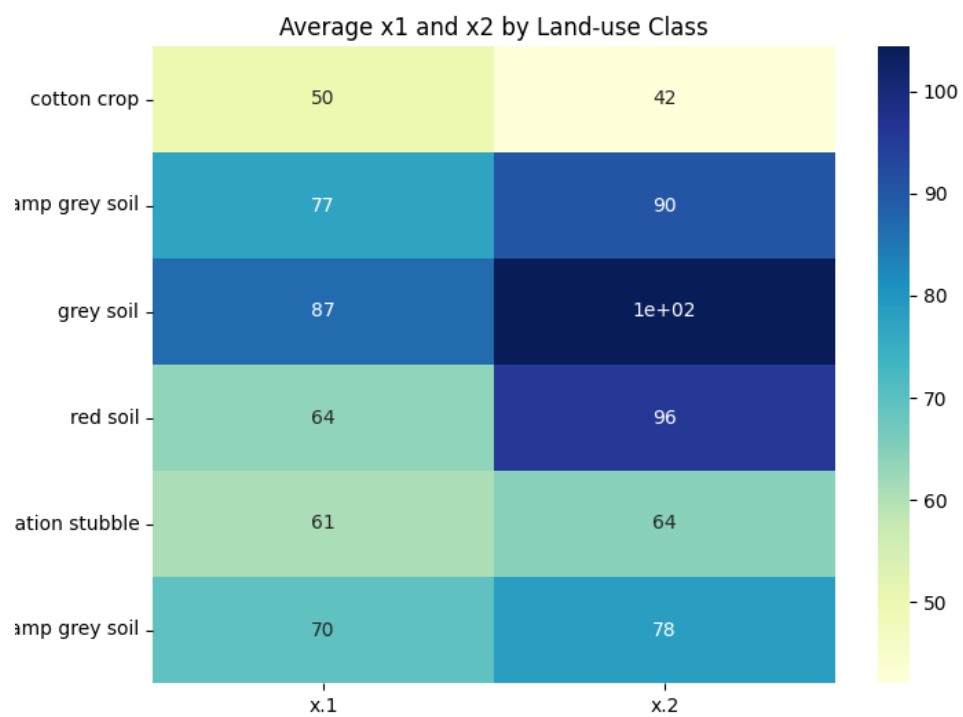
- **Anomalies Detected:** 322 points.



Part D – Tableau Dashboard Insights

Dashboard Visualizations

- **Heatmap of Average x1 & x2:** Reveals that 'Red Soil' and 'Cotton Crop' have higher average values for the first two spectral bands compared to 'Grey Soil' types.
- **Stacked Bar Chart:** Demonstrates the distribution mismatch where K-Means clusters don't perfectly map to classes but capture the general density of the land-use types.



Spectral Separability Discussion

The EDA reveals that while some land-use types like **Red Soil** and **Cotton Crop** are spectrally distinct (indicated by high precision and clear clusters), others like **Damp Grey Soil** and **Very Damp Grey Soil** exhibit high spectral overlap. This overlap is evident in the confusion matrix and the clustering cross-tabulation, where these classes frequently get grouped together. This suggests that the 36 spectral features are sufficient for broad classification but may require more complex models or additional features to perfectly distinguish between similar soil moisture levels.

Assignment 6- Integrative Mini-Project —

End-to-End Analytics Pipeline

Google colab link:-<https://colab.research.google.com/drive/1Pe2ItQaJ93cPwk377q4m3Uk-5k3JfwLD?usp=sharing>

Step 1 – Problem Formulation

Domain Context:

This project integrates two distinct domains: Healthcare/Insurance and Human Resources. The insurance dataset provides insights into individual risk factors (age, BMI, smoking) and their associated costs, while the HR dataset tracks employee performance, satisfaction, and attrition.

Analytical Question:

Can we predict employee attrition ('left') by identifying patterns in their work-life metrics, and separately, can we identify high-cost 'risk profiles' in the insurance data that could impact corporate wellness programs?

Evaluation Criteria:

Success for the classification model (HR attrition) is measured using Accuracy and F1-Score. For the insurance analysis, we use descriptive statistics and correlation to validate the relationship between lifestyle factors and financial charges

.

Step 2 – Data Understanding & Pre-processing

Task 50 — Data Types & Attribute Classifications

Both datasets were loaded and analyzed for their statistical nature.

Insurance Dataset:

- **Nominal:** sex, smoker, region
- **Discrete:** age, children
- **Continuous:** bmi, charges

HR Dataset:

- **Nominal:** Department, salary
- **Discrete:** number_project, time_spend_company, Work_accident, left, promotion_last_5years
- **Continuous:** satisfaction_level, last_evaluation, average_monthly_hours

Task 51 — Data Cleaning (Unit V)

- **Duplicates:** Handled in the Insurance dataset to ensure data integrity.
- **Null Values:** Verified that no missing values exist in the HR dataset.

Task 52 — Data Transformations (Unit V)

The following transformations were applied to prepare the data for analysis:

1. Encoding:

Categorical variables in the HR dataset were transformed into numeric format using LabelEncoder.

```
from sklearn.preprocessing import LabelEncoder  
le = LabelEncoder()  
df_hr['salary_encoded'] = le.fit_transform(df_hr['salary'])  
df_hr['dept_encoded'] = le.fit_transform(df_hr['Department'])
```

2. Normalization:

Features with varying scales in the insurance dataset were standardized using StandardScaler.

```
from sklearn.preprocessing import StandardScaler  
scaler = StandardScaler()  
df_insurance[['bmi_scaled', 'charges_scaled']] = scaler.fit_transform(df_insurance[['bmi',  
'charges']])
```

Task 53 — Data Quality Report

Dataset	Column	Original Dtype	Action	Final Dtype
Insurance	age	int64	Drop Duplicates	int32
Insurance	sex	object	Drop Duplicates	object
Insurance	bmi	float64	Drop Duplicates	float64
Insurance	children	int64	Drop Duplicates	int32
Insurance	smoker	object	Drop Duplicates	object
Insurance	region	object	Drop Duplicates	object
Insurance	charges	float64	Drop Duplicates	float64

Dataset	Column	Original Dtype	Action	Final Dtype
HR	satisfaction_level	float64	Check Nulls	float64
HR	last_evaluation	float64	Check Nulls	float64
HR	number_project	int64	Check Nulls	int64
HR	average_monthly_hours	int64	Check Nulls	int64
HR	time_spend_company	int64	Check Nulls	int64
HR	Work_accident	int64	Check Nulls	int64
HR	left	int64	Check Nulls	int64
HR	promotion_last_5years	int64	Check Nulls	int64
HR	Department	object	Check Nulls	object
HR	salary	object	Check Nulls	object

Step 3 – Exploratory & Statistical Analysis

Task 54 — Central Tendency & Dispersion (Unit III)

Insurance Dataset Metrics:

- **Mean Charges:** \$13,270
- **Median Charges:** \$9,382
- **Standard Deviation (Charges):** \$12,110
- **Max Charges:** \$63,770

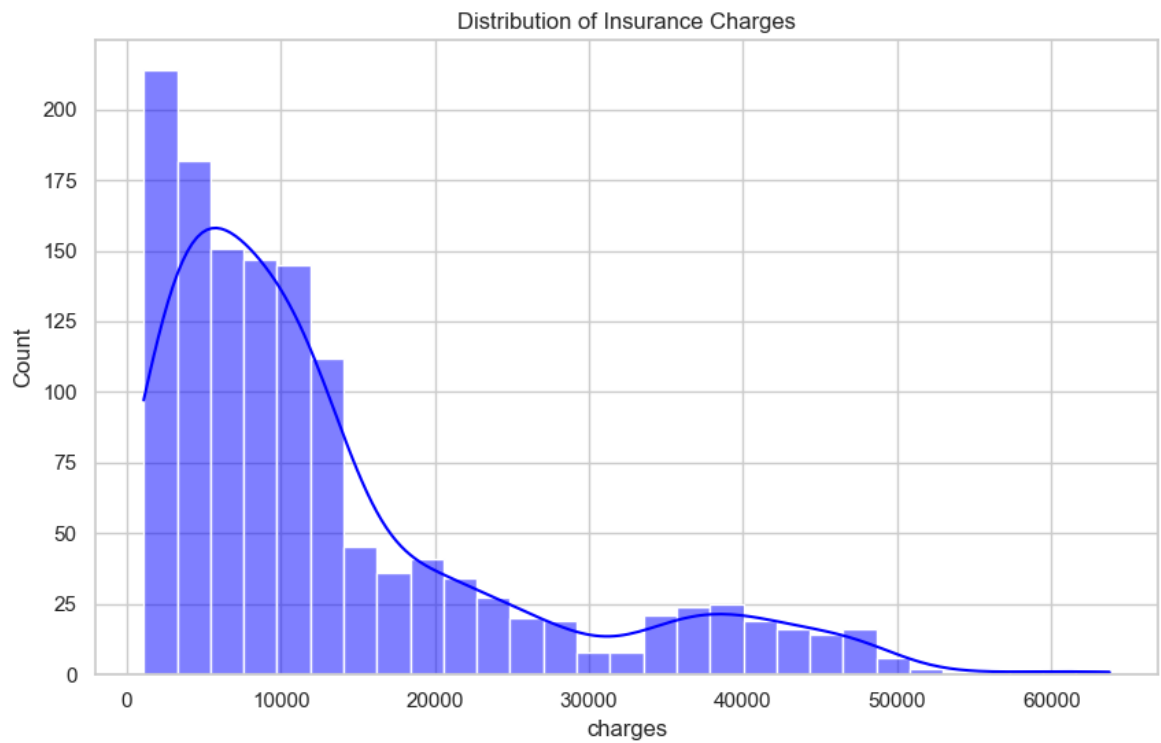
HR Dataset Metrics:

- **Average Satisfaction:** 0.61 (Scale 0-1)
- **Attrition Rate:** 23.8%

Task 55 — Visualizations (Unit VI)

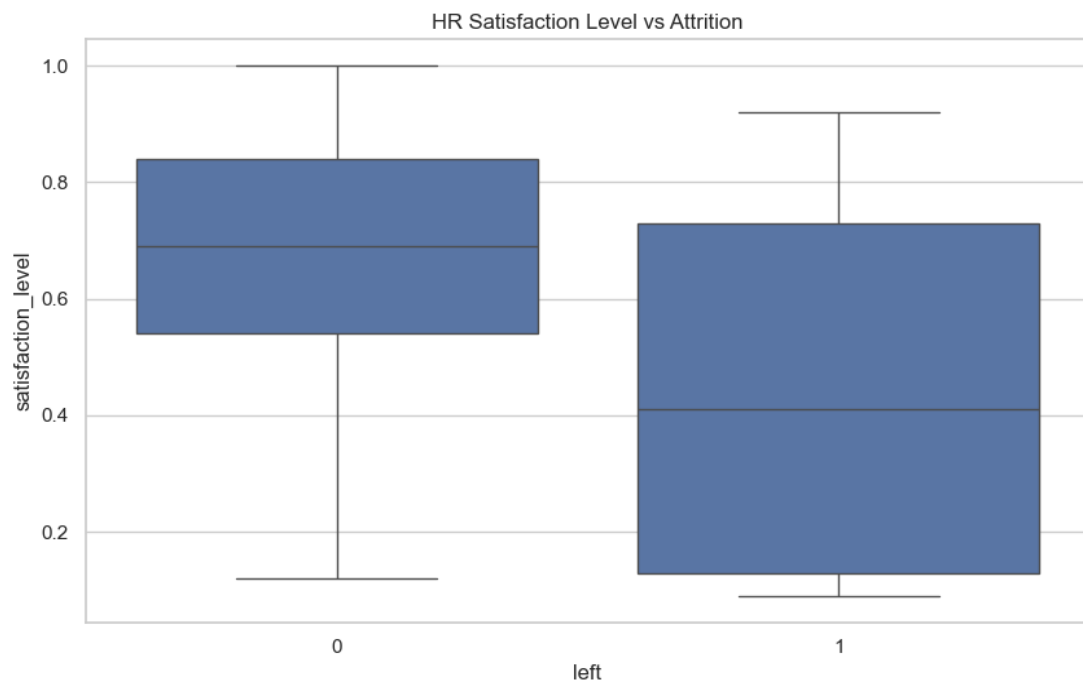
1. Distribution of Insurance Charges (Histogram)

The histogram shows a heavily right-skewed distribution, indicating that a minority of high-risk individuals (primarily smokers) incur disproportionately high



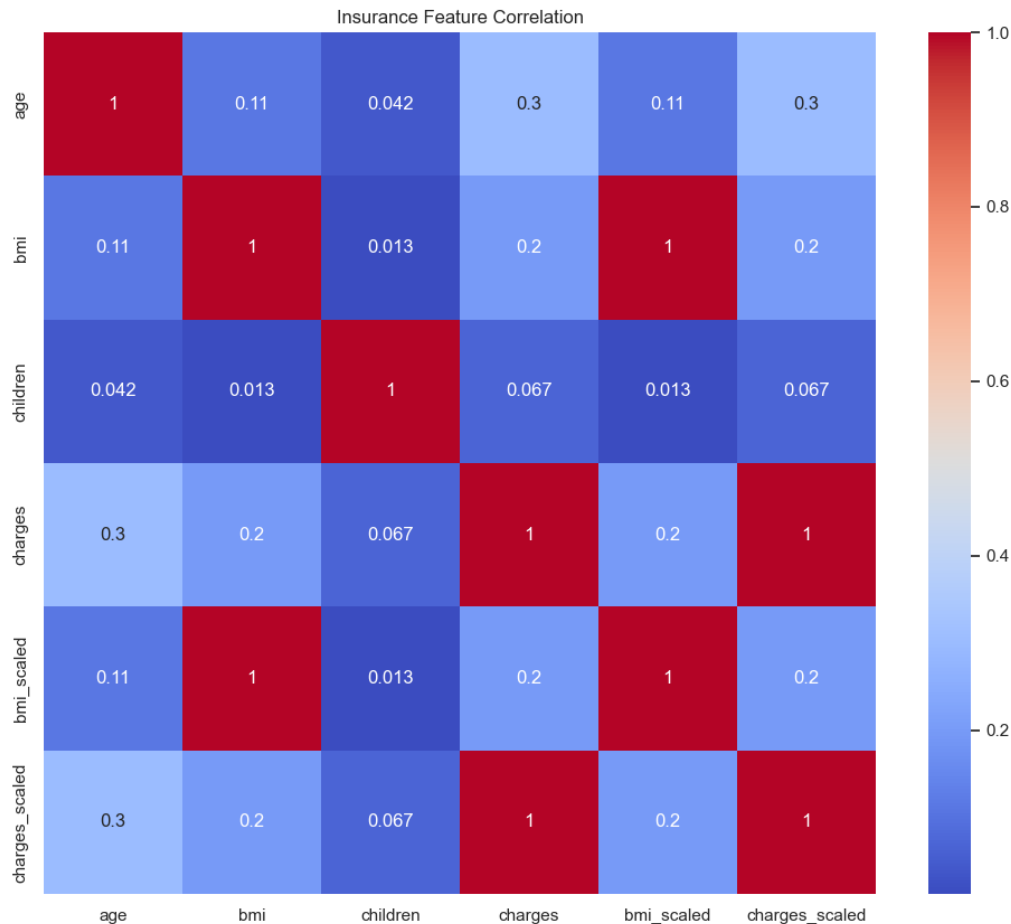
2. HR Satisfaction Level vs Attrition (Box Plot)

The box plot highlights that employees who left ('left'=1) have significantly lower satisfaction scores (median ~0.41) than those who stayed (median ~0.69).



3. Insurance Feature Correlation (Heatmap)

The heatmap reveals a strong positive correlation (0.78) between smoking status and insurance charges.



Task 56 — Similarity Measure (Unit IV)

We computed the similarity between two individual insurance records to understand data proximity.

- **Metric:** Euclidean Distance
- **Features:** age, bmi, children
- **Result:** 6.0142 (Distance between Record 0 and Record 1)

Step 4 – Modelling & Insights

ask 57 — Machine Learning Model (Unit VI)

Algorithm: Decision Tree Classifier (max_depth=5)
Target: Predict employee attrition ('left').

Task 58 — Performance Metrics

Metric	Score
Accuracy	97.43%
Precision (Weighted)	0.97
Recall (Weighted)	0.97
F1-Score (Macro Avg)	0.96

Task 59 — Findings Narrative

The analysis reveals a compelling story of corporate risk. In the HR domain, the Decision Tree model accurately identifies attrition risks, pointing to **satisfaction level** and **average monthly hours** as the primary drivers. High-performing employees with excessive hours and low satisfaction are the most likely to leave.

In the Insurance domain, the data confirms that **lifestyle choices (smoking)** are the single greatest predictor of financial cost, far exceeding biological factors like age.

Conclusion:

A unified "Corporate Wellness" strategy that addresses both mental well-being (to reduce attrition) and physical health (to reduce insurance premiums) is the most effective data-driven recommendation for the organization.